

LECTURE 3

Linear classification

Foundations of Artificial Intelligence

SEMTM0051 · 2026/27

Lecturer: Dr Roberto Colomboni



Today's lecture

Binary classification

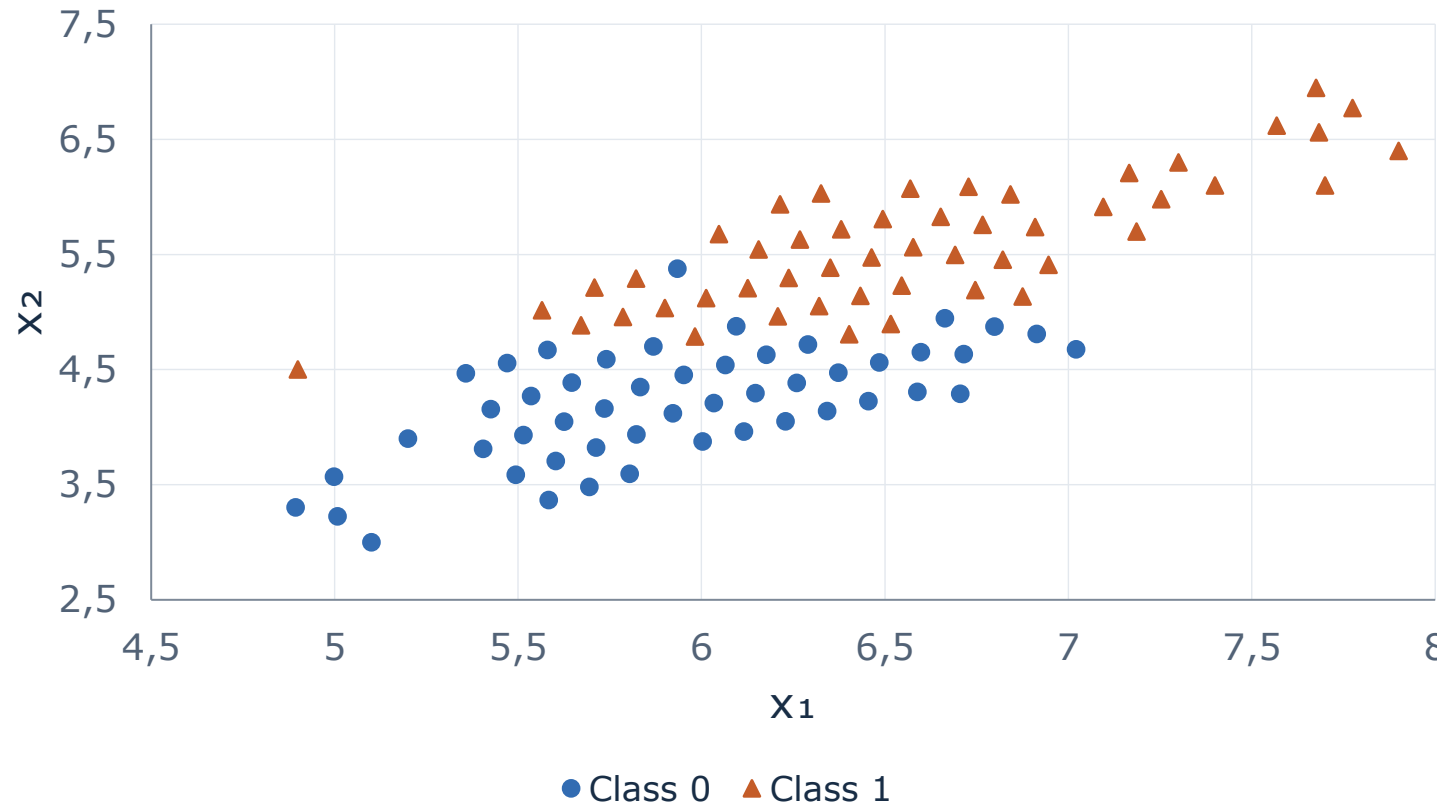
Classification error and logistic loss

Gradient descent

More than two classes

Binary classification

A synthetic dataset: 100 observations, two input features and two classes.



$$\mathbf{x}_i = (x_{i1}, x_{i2}), \quad y_i \in \{0,1\}$$

x_1 : first input feature

x_2 : second input feature

Class 0: 50 observations

Class 1: 50 observations

Learn a rule to predict the class of an input.

A modified Iris dataset: 100 observations, two features and two classes.

Dataset and instructions in the folder: Week 3 - Examples

Measuring classification error

A regression **predictor** f
returns a number

Regression

$$\ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2$$

Empirical squared loss

$$\mathcal{L}_{\text{sq}}(f) = \sum_{i=1}^N \ell_{\text{sq}}(f(\mathbf{x}_i), y_i)$$

Larger numerical errors
receive larger penalties

A **classifier** h
returns a class label.

Classification

$$\ell_{0,1}(\hat{y}, y) = \begin{cases} 0, & \hat{y} = y \\ 1, & \hat{y} \neq y \end{cases}$$

Number of mistakes

$$\mathcal{E}(h) = \sum_{i=1}^N \ell_{0,1}(h(\mathbf{x}_i), y_i)$$

All mistakes are penalized
the same way

Predicting a probability first

A label gives a decision.

A probability also records how strongly the model favours that label.

$$p: \mathbb{R}^d \rightarrow [0,1]$$

Probability of class 1

$$p(\mathbf{x})$$

Probability of class 0

$$1 - p(\mathbf{x})$$

Example: if $p(\mathbf{x}) = 0.51$ we think that there is

- a 0.51 probability that the correct class of input $\mathbf{x}$ is 1
- a 0.49 probability that the correct class of input $\mathbf{x}$ is 0

If we must choose one label, which should we predict?

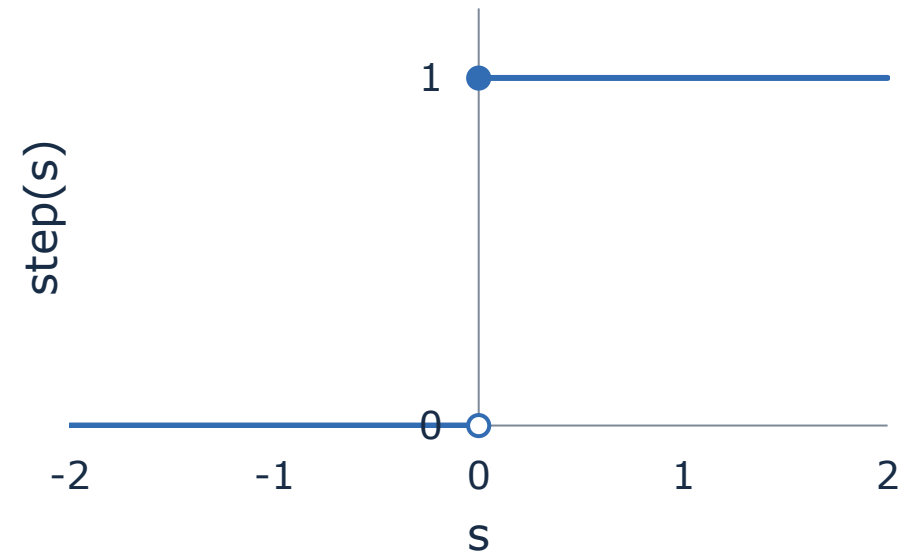
The plug-in classification rule

Choose the label with the larger probability. At a tie, we choose class 1.

$$h_p(\mathbf{x}) = \begin{cases} 0, & p(\mathbf{x}) < 1/2 \\ 1, & p(\mathbf{x}) \geq 1/2 \end{cases}$$

Heaviside step function

$$\text{step}(s) := \begin{cases} 0, & s < 0, \\ 1, & s \geq 0. \end{cases}$$



$$h_p(\mathbf{x}) = \text{step}\left(p(\mathbf{x}) - \frac{1}{2}\right)$$

How do we choose the probability function?

How do we build probability predictions?

Regression suggests using an intercept and one weight for each input feature.

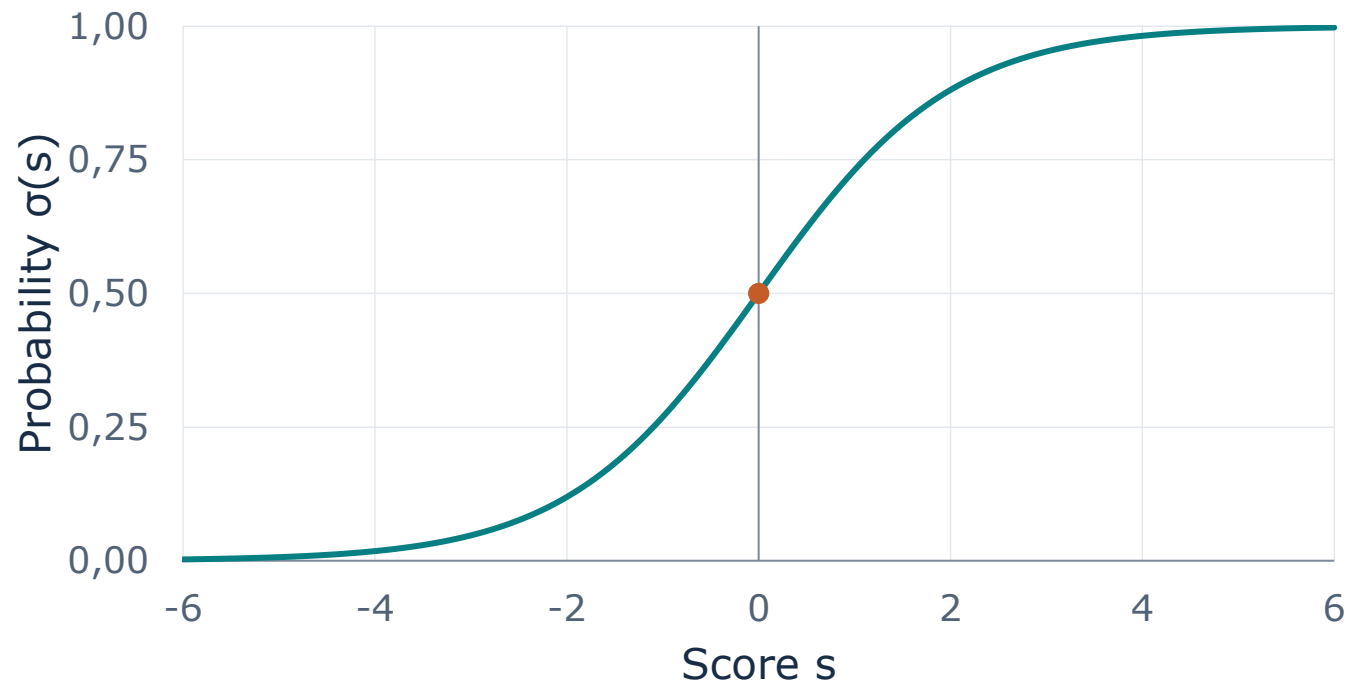
$$f_{\boldsymbol{\theta}}(\mathbf{x}) = b + \mathbf{x}\mathbf{w} = b + \sum_{j=1}^d x_j w_j$$

$$\begin{aligned} b &\in \mathbb{R}, & \mathbf{w} &= (w_1, \dots, w_d)^\top \in \mathbb{R}^{d \times 1} \\ \boldsymbol{\theta} &= \begin{pmatrix} b \\ \mathbf{w} \end{pmatrix} = (b, w_1, \dots, w_d)^\top \in \mathbb{R}^{(d+1) \times 1} \end{aligned} \quad \begin{aligned} d = 1: & \quad f_{\boldsymbol{\theta}}(\mathbf{x}) = b + x_1 w_1 \\ d = 2: & \quad f_{\boldsymbol{\theta}}(\mathbf{x}) = b + x_1 w_1 + x_2 w_2 \\ d = 3: & \quad f_{\boldsymbol{\theta}}(\mathbf{x}) = b + x_1 w_1 + x_2 w_2 + x_3 w_3 \end{aligned}$$

However, the score can be **any real number**!
It cannot serve directly as a probability.

The sigmoid function

A **smooth, increasing function** turns each real score into a number between zero and one.



$$\sigma(s) := \frac{1}{1 + e^{-s}} = \frac{e^s}{1 + e^s}$$

$$0 < \sigma(s) < 1$$

$$\sigma(0) = \frac{1}{2}$$

$s < 0$ favours class 0.

$s > 0$ favours class 1.

Probabilities, labels and the boundary

The sigmoid is increasing and maps score zero to probability one half.

$$p_{\theta}(\mathbf{x}) = \sigma(f_{\theta}(\mathbf{x})) = \sigma(b + \mathbf{xw})$$

$$h_{p_{\theta}}(\mathbf{x}) = 1 \quad \Leftrightarrow \quad p_{\theta}(\mathbf{x}) \geq \frac{1}{2} \quad \Leftrightarrow \quad f_{\theta}(\mathbf{x}) \geq 0 \quad \Leftrightarrow \quad b + \mathbf{xw} \geq 0$$

Separating hyperplane: $b + \mathbf{xw} = 0$

The boundary is a hyperplane with d features, provided the weight vector $\mathbf{w}$ is nonzero.

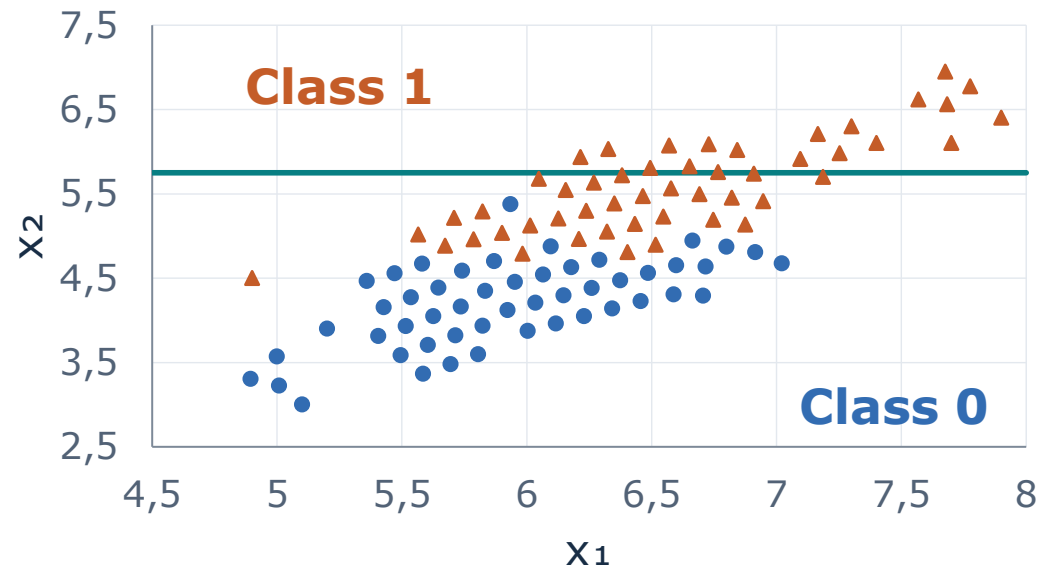
If $d = 1$, $b + x_1w_1 = 0$ determines a point of the Euclidean line

If $d = 2$, $b + x_1w_1 + x_2w_2 = 0$ determines a straight line of the Euclidean plane

If $d = 3$, $b + x_1w_1 + x_2w_2 + x_3w_3 = 0$ determines a plane of the Euclidean space

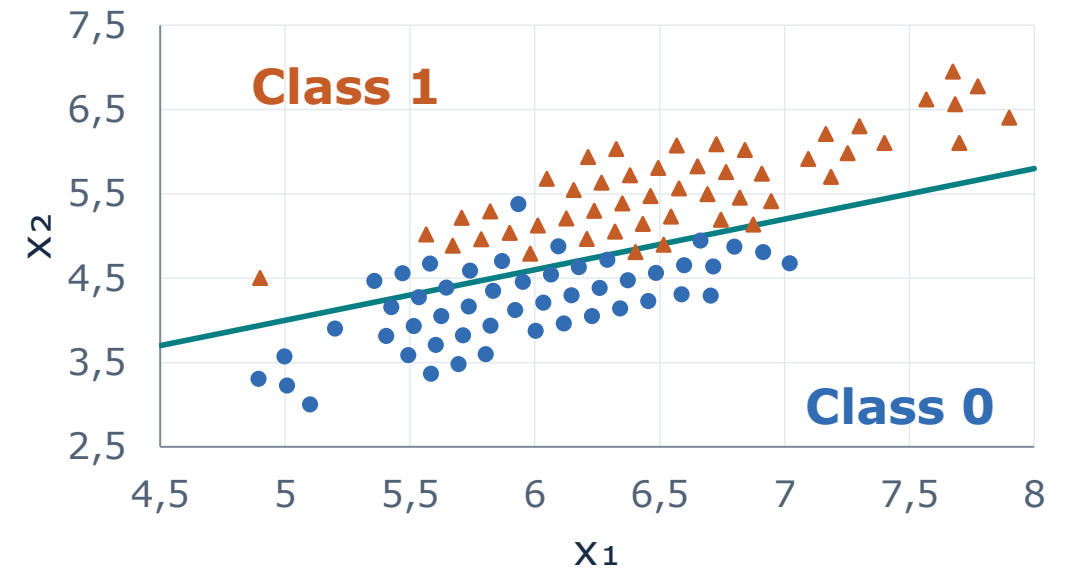
Comparing two candidate boundaries

The same observations can receive different labels when we change the parameters.



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.75$$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 0.6x_1 + 1$$

The classifier on the right makes fewer mistakes on these observations

Finding the parameters

We want the smallest mistake count.
Finding the parameter θ^* that minimizes

$$\theta \mapsto \mathcal{E}(\theta) = \sum_{i=1}^N \ell_{0,1}(h_{p_\theta}(\mathbf{x}_i), y_i)$$

is a **combinatorial** problem.

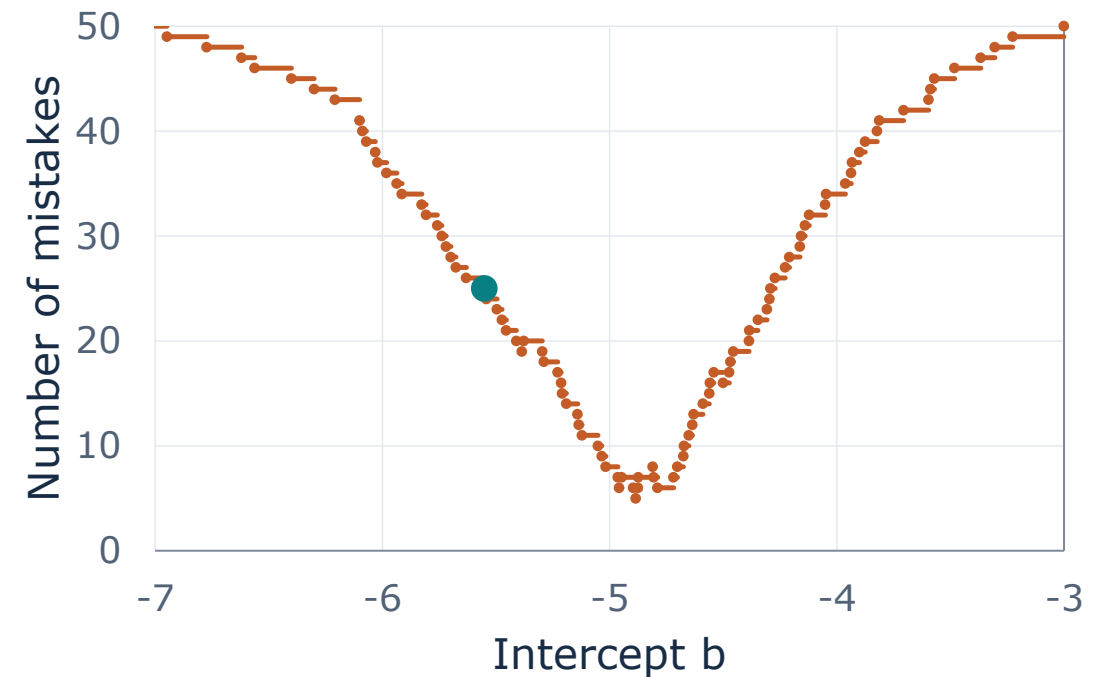
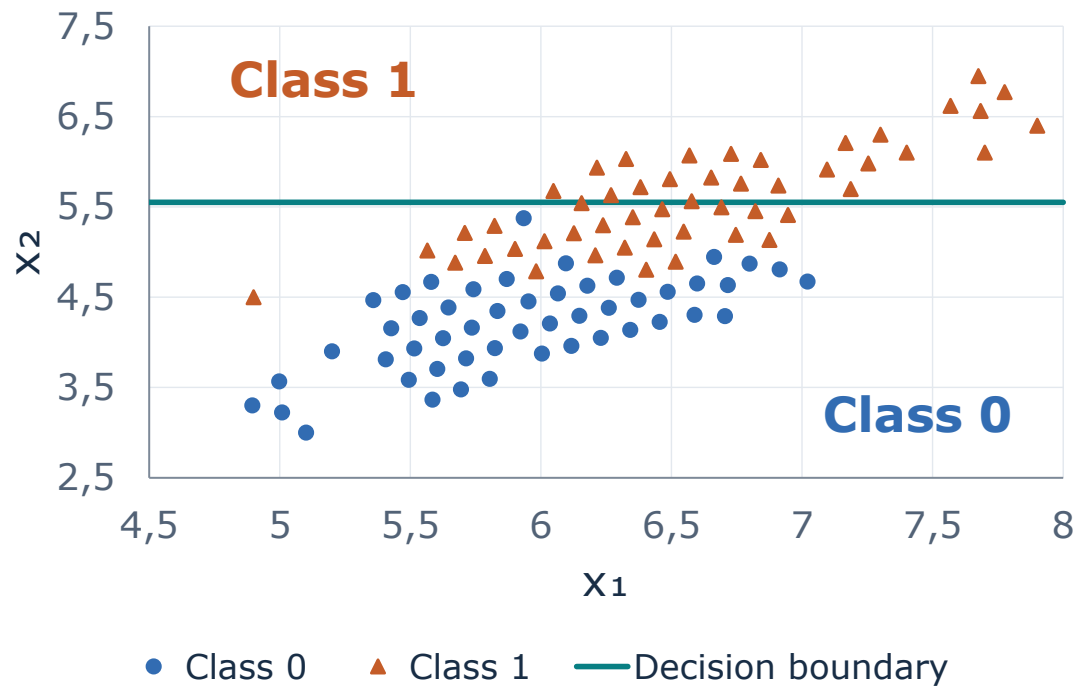
Can we compute an **approximate solution** via
gradient descent as in regression?

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \mathcal{E}(\theta^{(t)}), \quad \eta > 0$$

What happens to the mistake count when we move the boundary a little?

Moving the boundary and counting mistakes

Fix the weights and vary only the intercept. The line moves parallel to itself.



$$b = -5.55, \quad w_1 = 0, \quad w_2 = 1$$

The count changes only when the line crosses an observed point.

Flat regions give no direction

If every observed score is **nonzero**
small parameter changes preserve all predicted labels.

$$\mathcal{E}(\boldsymbol{\theta} + \Delta) = \mathcal{E}(\boldsymbol{\theta}) \quad \text{for sufficiently small } \Delta$$

$$\nabla \mathcal{E}(\boldsymbol{\theta}) = \mathbf{0}$$

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{E}(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)}$$

Gradient descent will **not move!!!**

What does the mistake count miss?

Suppose the observed class is 0. Both predictions below choose class 1.

	Prediction A	Prediction B
Probability of class 1	0.51	0.90
Predicted class	1	1
Zero-one loss	1	1
Probability of the observed class	0.49	0.10

At 0.51, the model is almost guessing: we want a milder penalty.

At 0.90, it is confidently wrong: we want a larger penalty.

Rather than scoring only the predicted label,
we penalise assigning low probability to the observed class.

A smooth loss for fitting probabilities

We take the negative logarithm of the probability assigned to the observed class.

$$\ell_{\log}(p, y) = \begin{cases} -\log(p), & y = 1, \\ -\log(1 - p), & y = 0. \end{cases}$$

$$\ell_{\log}(p, y) = -y\log(p) - (1 - y)\log(1 - p)$$

Wrong, but uncertain

$$\begin{aligned} \ell_{\log}(0.51, 0) &= -\log(0.49) \\ &\approx 0.713 \end{aligned}$$

Confidently wrong

$$\begin{aligned} \ell_{\log}(0.90, 0) &= -\log(0.10) \\ &\approx 2.303 \end{aligned}$$

Logistic loss is also called binary cross-entropy. We use natural logarithms.

The empirical logistic loss

The observations stay fixed.
The probability predictions depend on the parameters.

$$p_i = p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \sigma(b + \mathbf{x}_i \mathbf{w})$$

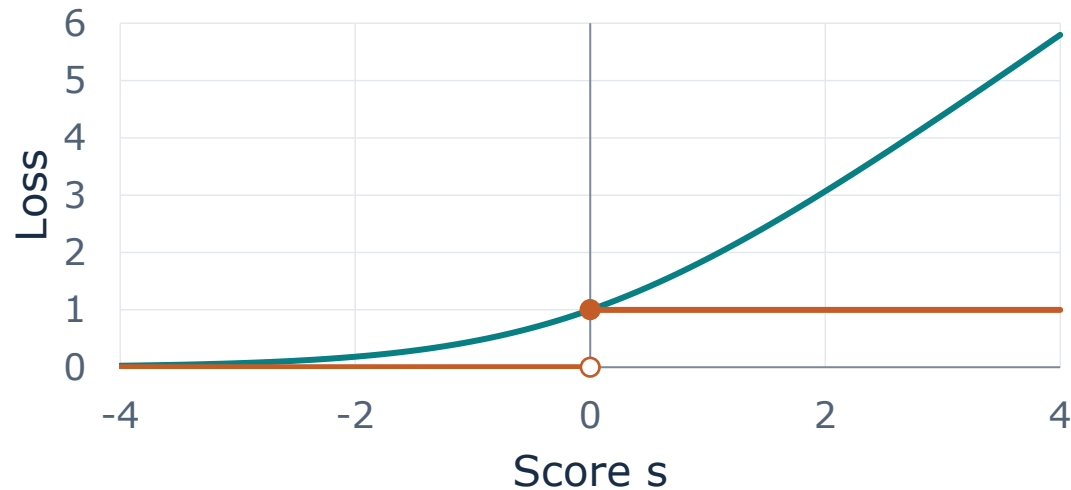
$$\mathcal{L}_{\log}(\boldsymbol{\theta}) = \sum_{i=1}^N \ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i)$$

**We fit this smooth objective,
while the mistake count $\varepsilon(\boldsymbol{\theta})$ remains our classification measure.**

Comparing the two losses

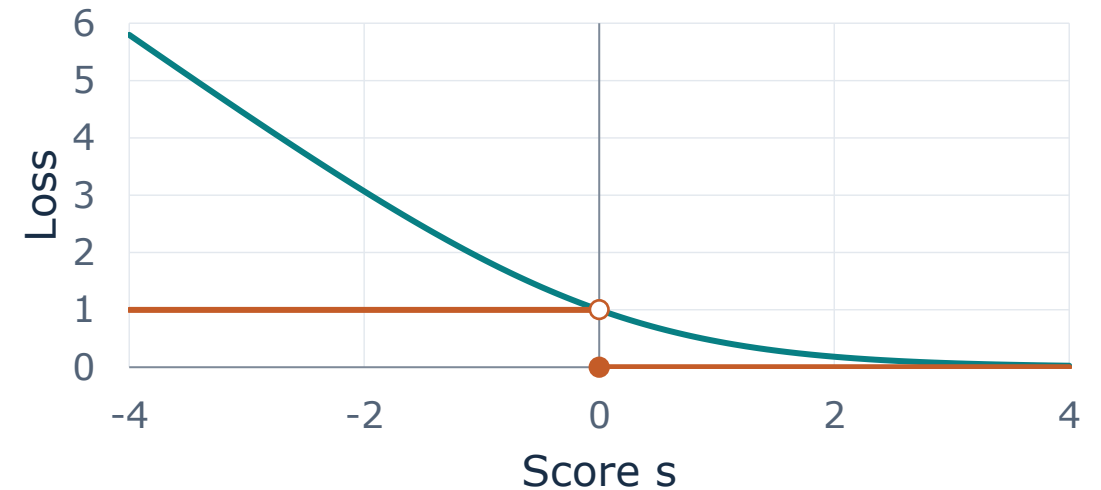
A mistake gives the observed class probability at most $1/2$,
so its logistic loss is at least $\log 2$.

Observed label $y = 0$



— Logistic loss / log 2 — Zero-one loss

Observed label $y = 1$



— Logistic loss / log 2 — Zero-one loss

$$\ell_{0,1}(\text{step}(s), y) \leq \frac{\ell_{\log}(\sigma(s), y)}{\log 2}$$

Bounding the number of mistakes

The inequality holds for each observation

We can therefore add it over the dataset

$$\begin{aligned}\mathcal{E}(\boldsymbol{\theta}) &= \sum_{i=1}^N \ell_{0,1}(h_{p_{\boldsymbol{\theta}}}(\mathbf{x}_i), y_i) \\ &= \sum_{i=1}^N \ell_{0,1}(\text{step}(f_{\boldsymbol{\theta}}(\mathbf{x}_i)), y_i) \leq \sum_{i=1}^N \frac{\ell_{\log}(\sigma(f_{\boldsymbol{\theta}}(\mathbf{x}_i)), y_i)}{\log 2} \\ &= \frac{1}{\log(2)} \sum_{i=1}^N \ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = \frac{1}{\log(2)} \cdot \mathcal{L}_{\log}(\boldsymbol{\theta})\end{aligned}$$

A control over the empirical logistic loss
gives a control on the number of mistakes

The design matrix and the parameters

As in Lecture 2, a first column of ones includes the intercept.

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{Nd} \end{pmatrix} \quad \boldsymbol{\theta} = \begin{pmatrix} b \\ w_1 \\ \vdots \\ w_d \end{pmatrix}$$

$$\mathbf{X} \in \mathbb{R}^{N \times (d+1)}, \quad \boldsymbol{\theta} \in \mathbb{R}^{(d+1) \times 1}$$

$$\theta_1 = b, \quad \theta_{k+1} = w_k, \quad k = 1, \dots, d$$

The first coordinate is the intercept. The remaining coordinates are the weights.

The three functions

Fix one input and its observed label. We vary the model parameters.

Affine score

$$F_i(\boldsymbol{\theta}) := f_{\boldsymbol{\theta}}(\mathbf{x}_i) = b + \mathbf{x}_i \mathbf{w} = (1, \mathbf{x}_i) \boldsymbol{\theta} = \sum_{r=1}^{d+1} X_{ir} \theta_r$$

Sigmoid

$$\sigma(s) = \frac{1}{1 + e^{-s}} \quad (s \in \mathbb{R})$$

Logistic loss, with the observed label fixed

$$G_i(p) := \ell_{\log}(p, y_i) = -y_i \log p - (1 - y_i) \log(1 - p)$$

The chain rule for one observation

The loss for one observation is the composition of these three functions.

$$\ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = (G_i \circ \sigma \circ F_i)(\boldsymbol{\theta}) = G_i \left(\sigma(F_i(\boldsymbol{\theta})) \right)$$

To compute the partial derivatives with respect to θ_j of this quantity
We apply the **chain rule to this composition.**

$$\begin{aligned} \frac{\partial}{\partial \theta_j} (G_i \circ \sigma \circ F_i)(\boldsymbol{\theta}) &= G'_i \left(\sigma(F_i(\boldsymbol{\theta})) \right) \cdot \sigma'(F_i(\boldsymbol{\theta})) \cdot \frac{\partial F_i}{\partial \theta_j}(\boldsymbol{\theta}) \\ &= \underbrace{\frac{\partial \ell_{\log}}{\partial p} \left(\sigma(f_{\boldsymbol{\theta}}(\mathbf{x}_i)), y_i \right)}_{\text{first term}} \cdot \underbrace{\sigma'(f_{\boldsymbol{\theta}}(\mathbf{x}_i))}_{\text{second term}} \cdot \underbrace{\frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j}}_{\text{third term}} \end{aligned}$$

First term: Differentiating the logistic loss

Hold the observed label fixed and vary the probability.

$$\ell_{\log}(p, y) = -y \log p - (1 - y) \log(1 - p)$$

$$\begin{aligned} \frac{\partial \ell_{\log}}{\partial p}(p, y) &= -\frac{y}{p} + \frac{1 - y}{1 - p} = \frac{-y(1 - p) + (1 - y)p}{p(1 - p)} \\ &= \frac{-y + yp + p - yp}{p(1 - p)} = \frac{p - y}{p(1 - p)} \end{aligned}$$

$$\boxed{\frac{\partial \ell_{\log}}{\partial p}(\sigma(f_{\theta}(\mathbf{x}_i)), y_i) = \frac{\sigma(f_{\theta}(\mathbf{x}_i)) - y_i}{\sigma(f_{\theta}(\mathbf{x}_i))(1 - \sigma(f_{\theta}(\mathbf{x}_i)))} = \frac{p_{\theta}(\mathbf{x}_i) - y_i}{p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}}$$

Second term: Differentiating the sigmoid

Write the sigmoid as a composition and apply the chain rule again.

Inner function u

$$u(s) = 1 + e^{-s}, \quad u'(s) = -e^{-s}$$

Outer function v

$$v(z) = \frac{1}{z}, \quad v'(z) = -\frac{1}{z^2}$$

$$\sigma(s) = \frac{1}{1 + e^{-s}} = v(u(s))$$

$$\sigma'(s) = v'(u(s)) u'(s) = -\frac{1}{(1 + e^{-s})^2} (-e^{-s}) = \frac{e^{-s}}{(1 + e^{-s})^2}$$

$$= \frac{1}{1 + e^{-s}} \cdot \frac{e^{-s}}{1 + e^{-s}} = \frac{1}{1 + e^{-s}} \cdot \left(1 - \frac{1}{1 + e^{-s}}\right) = \sigma(s)(1 - \sigma(s))$$

$$\boxed{\sigma'(f_{\theta}(\mathbf{x}_i)) = \sigma(f_{\theta}(\mathbf{x}_i)) (1 - \sigma(f_{\theta}(\mathbf{x}_i))) = p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}$$

Third term: Differentiating the affine score

The inputs are fixed, so only the matching parameter contributes.

The affine score

$$f_{\boldsymbol{\theta}}(\mathbf{x}_i) = \sum_{r=1}^{d+1} X_{ir} \theta_r$$

The Kronecker delta

$$\frac{\partial \theta_r}{\partial \theta_j} = \delta_{rj} = \begin{cases} 1, & r = j, \\ 0, & r \neq j. \end{cases}$$

$$\frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \sum_{r=1}^{d+1} X_{ir} \theta_r = \sum_{r=1}^{d+1} X_{ir} \frac{\partial \theta_r}{\partial \theta_j} = \sum_{r=1}^{d+1} X_{ir} \delta_{rj} = X_{ij}$$

$$\boxed{\frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j} = X_{ij}} \quad j = 1, \dots, d + 1$$

Putting the three terms together

Substitute the first, second and third terms into the chain rule.

$$\begin{aligned} & \frac{\partial}{\partial \theta_j} \ell_{\log}(p_{\theta}(\mathbf{x}_i), y_i) \\ &= \underbrace{\frac{\partial \ell_{\log}}{\partial p}(\sigma(f_{\theta}(\mathbf{x}_i)), y_i)}_{\text{first term}} \cdot \underbrace{\sigma'(f_{\theta}(\mathbf{x}_i))}_{\text{second term}} \cdot \underbrace{\frac{\partial f_{\theta}(\mathbf{x}_i)}{\partial \theta_j}}_{\text{third term}} \\ &= \underbrace{\frac{p_{\theta}(\mathbf{x}_i) - y_i}{p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}}_{\text{first term}} \cdot \underbrace{p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}_{\text{second term}} \cdot \underbrace{X_{ij}}_{\text{third term}} \\ &= (p_{\theta}(\mathbf{x}_i) - y_i)X_{ij} \end{aligned}$$

Empirical gradient and gradient descent

The empirical loss is a sum, so we add the observation derivatives.

$$\frac{\partial \mathcal{L}_{\log}}{\partial \theta_j}(\boldsymbol{\theta}) = \sum_{i=1}^N \frac{\partial}{\partial \theta_j} \ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = \sum_{i=1}^N X_{ij}(p_{\boldsymbol{\theta}}(\mathbf{x}_i) - y_i)$$

Sigmoid applied to each entry

$$\boldsymbol{\sigma}((s_1, \dots, s_N)^\top) := (\sigma(s_1), \dots, \sigma(s_N))^\top$$

One probability per observation

$$\mathbf{p}_{\boldsymbol{\theta}} := (p_{\boldsymbol{\theta}}(\mathbf{x}_1), \dots, p_{\boldsymbol{\theta}}(\mathbf{x}_N))^\top$$

$$\mathbf{p}_{\boldsymbol{\theta}} = \boldsymbol{\sigma}(\mathbf{X}\boldsymbol{\theta}), \quad \mathbf{y} = (y_1, \dots, y_N)^\top$$

Gradient

$$\nabla \mathcal{L}_{\log}(\boldsymbol{\theta}) = \mathbf{X}^\top(\mathbf{p}_{\boldsymbol{\theta}} - \mathbf{y}) = \mathbf{X}^\top(\boldsymbol{\sigma}(\mathbf{X}\boldsymbol{\theta}) - \mathbf{y})$$

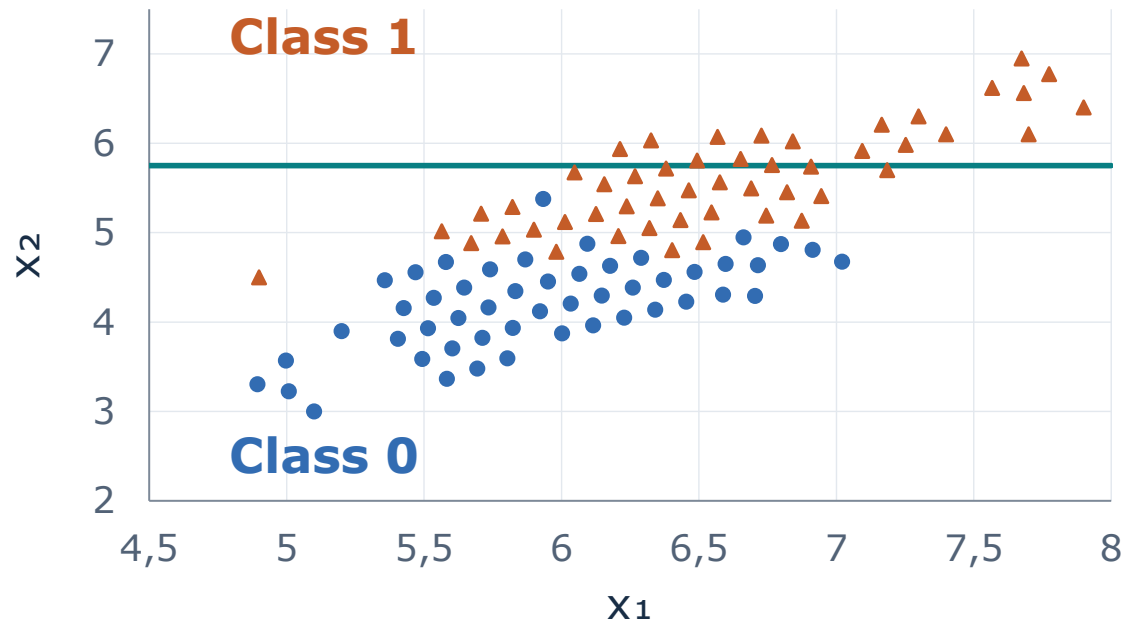
Gradient descent

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}_{\log}(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)} - \eta \mathbf{X}^\top(\boldsymbol{\sigma}(\mathbf{X}\boldsymbol{\theta}^{(t)}) - \mathbf{y})$$

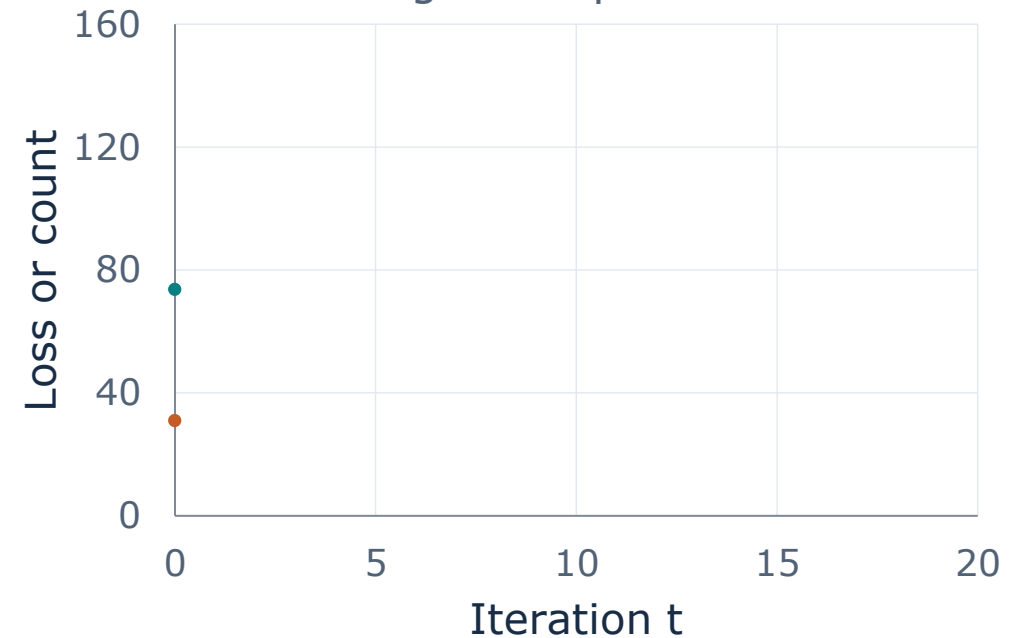
Gradient descent: $\eta = 0.001$, $t = 0$

Initial coefficients: $b^{(0)} = -5.75$, $w_1^{(0)} = 0$, $w_2^{(0)} = 1$

Learning rate: $\eta = 0.001$



● Class 0 ▲ Class 1 — Decision boundary



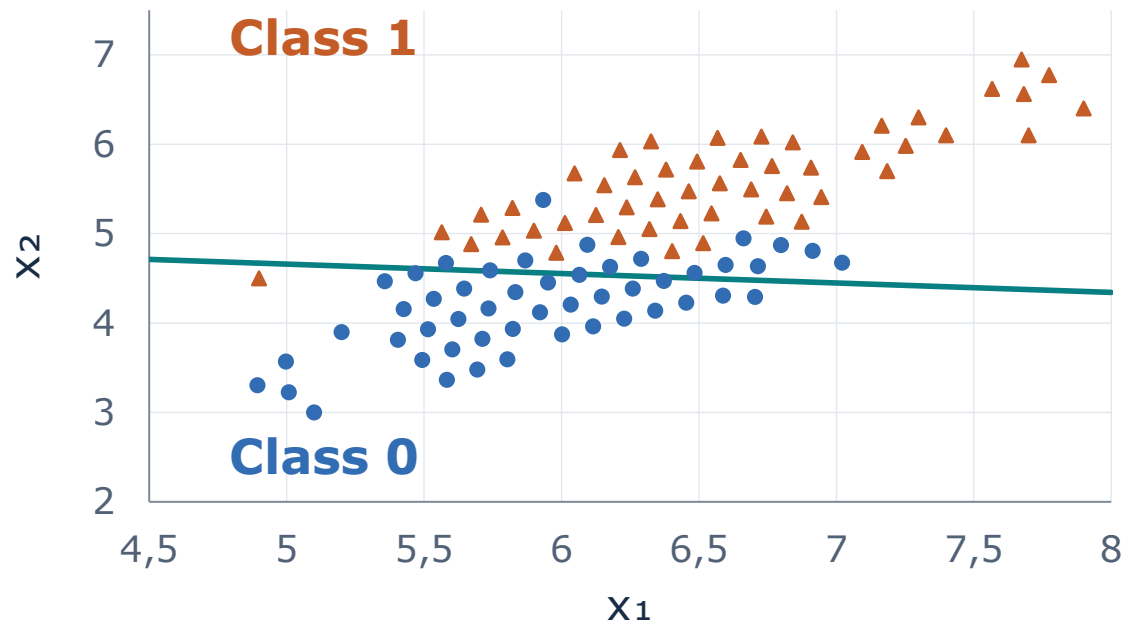
—●— Logistic loss / log 2 —●— Number of mistakes

$$x_2 = 5.7500 + 0.0000x_1$$

Logistic loss: 51.048
Mistakes: 31 of 100

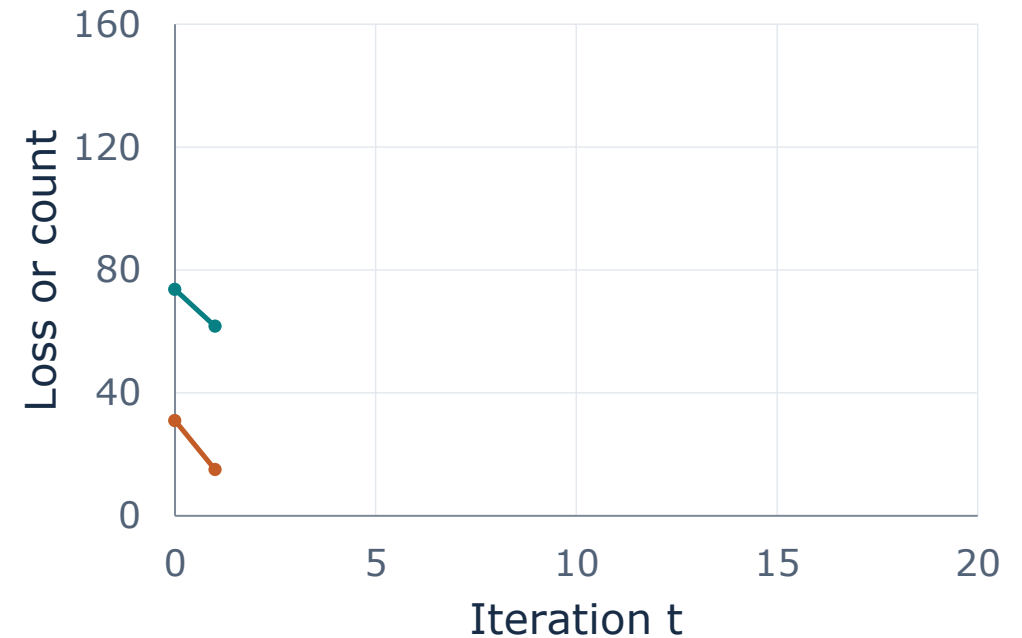
Gradient descent: $\eta = 0.001$, $t = 1$

Coefficients: $b^{(1)} \approx -5.732582$ $w_1^{(1)} \approx 0.116619$ $w_2^{(1)} \approx 1.105206$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.1869 - 0.1055x_1$$

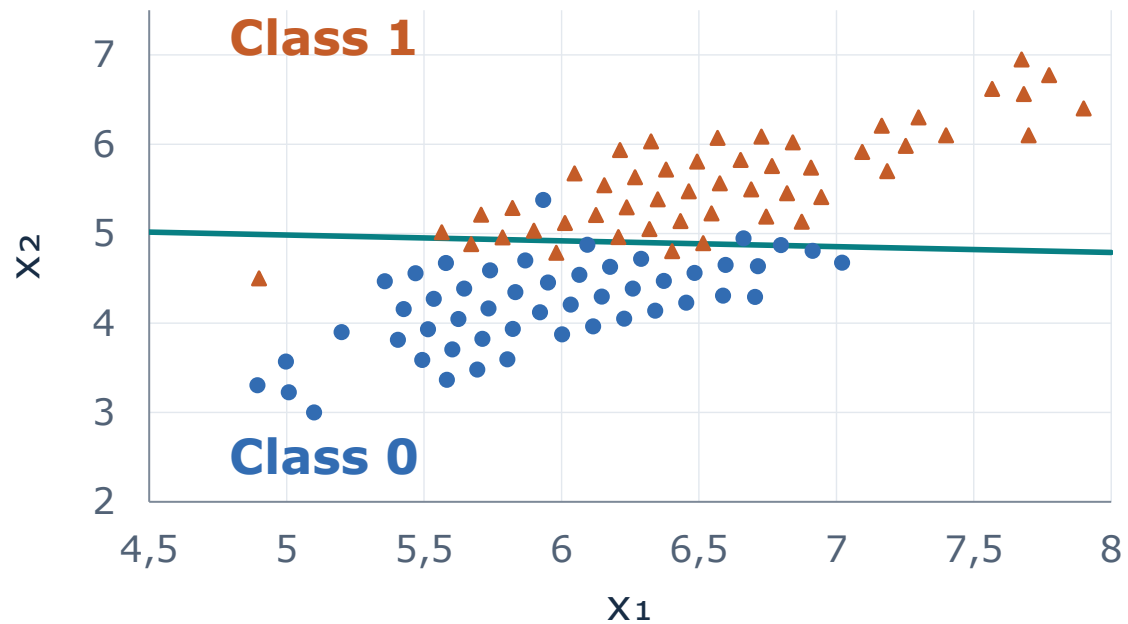


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 42.733
Mistakes: 15 of 100

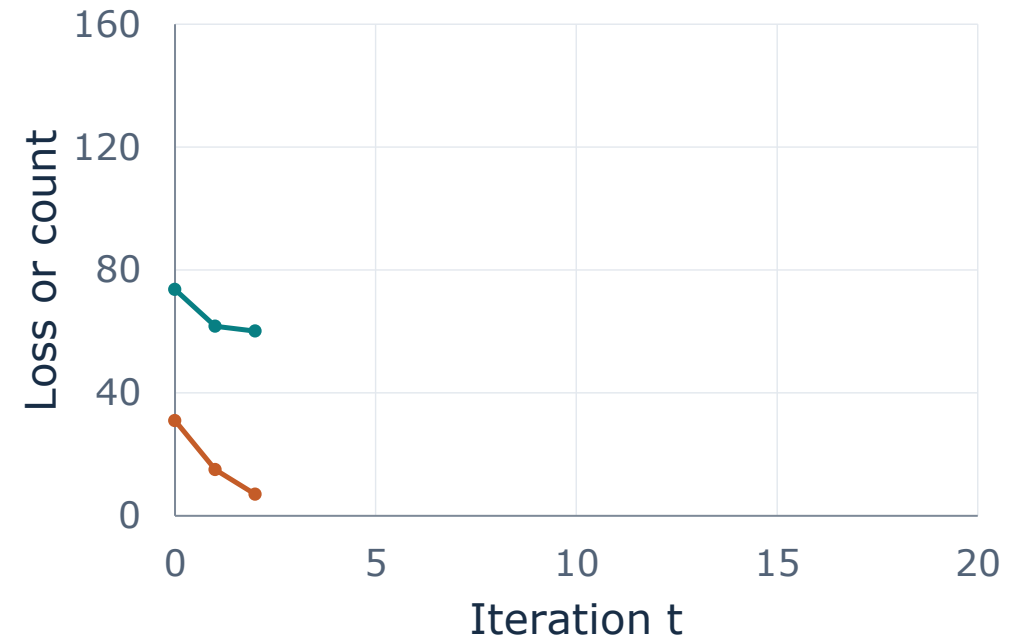
Gradient descent: $\eta = 0.001$, $t = 2$

Coefficients: $b^{(2)} \approx -5.740878$ $w_1^{(2)} \approx 0.070300$ $w_2^{(2)} \approx 1.081132$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.3101 - 0.0650x_1$$

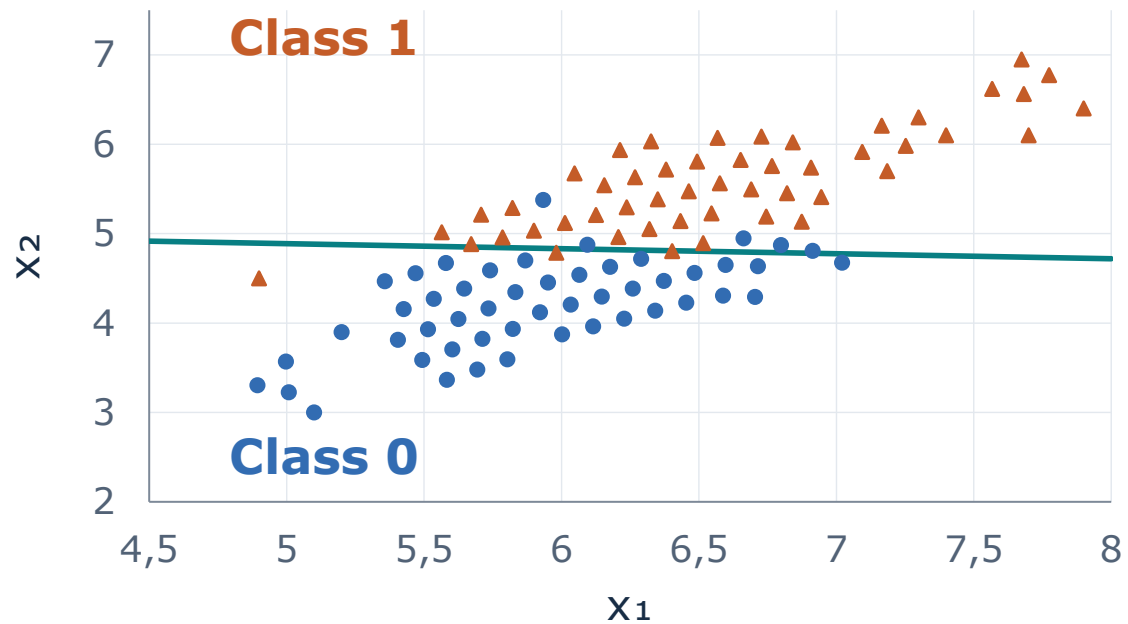


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 41.668
Mistakes: 7 of 100

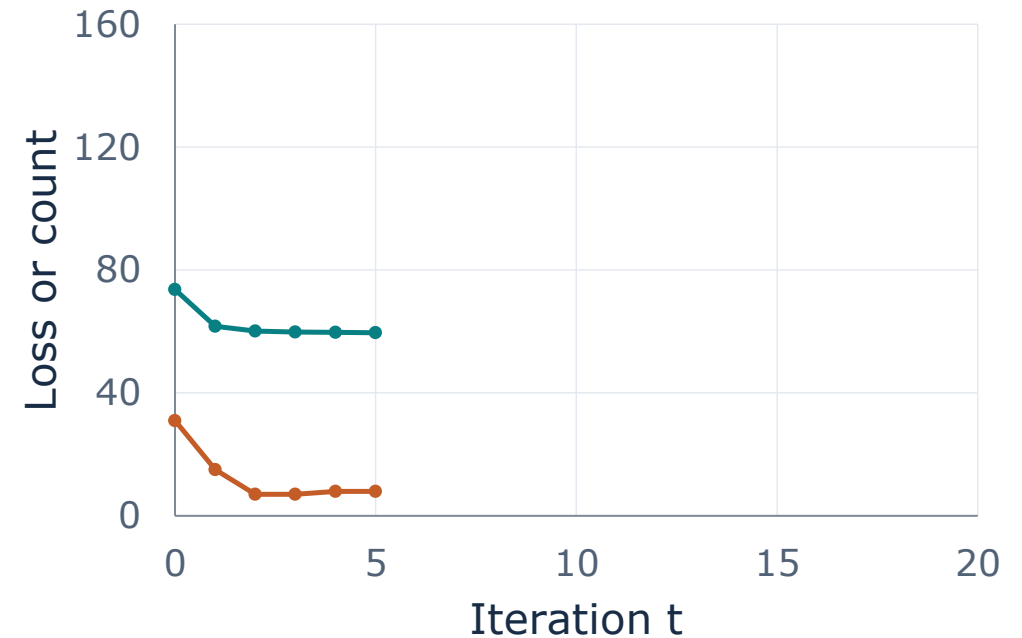
Gradient descent: $\eta = 0.001$, $t = 5$

Coefficients: $b^{(5)} \approx -5.744795$ $w_1^{(5)} \approx 0.062560$ $w_2^{(5)} \approx 1.111266$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.1696 - 0.0563x_1$$

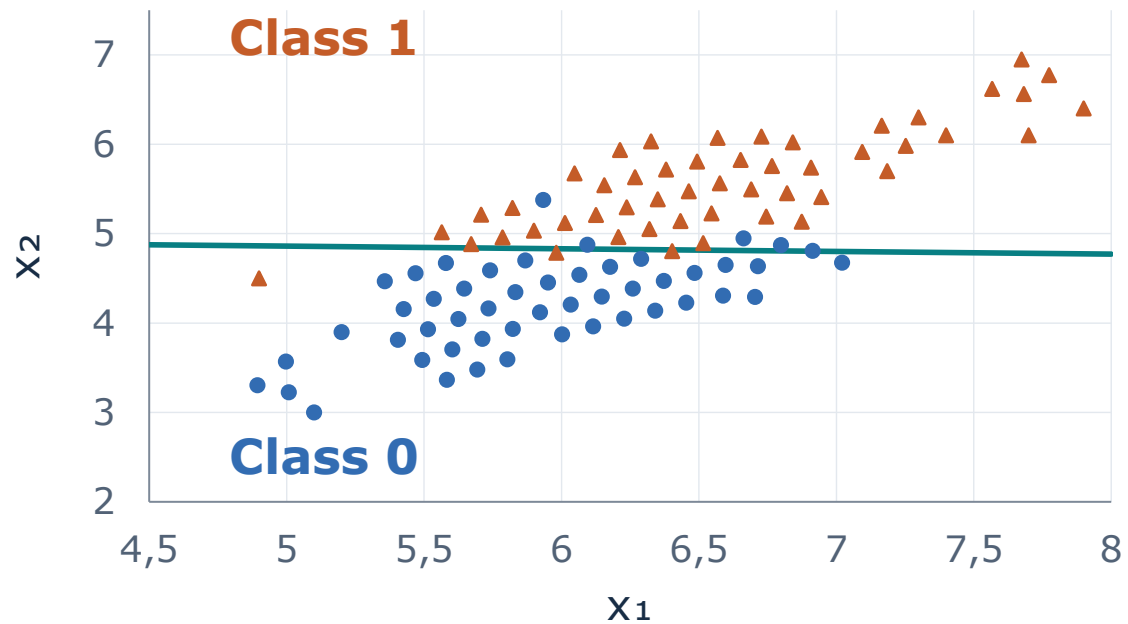


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 41.300
Mistakes: 8 of 100

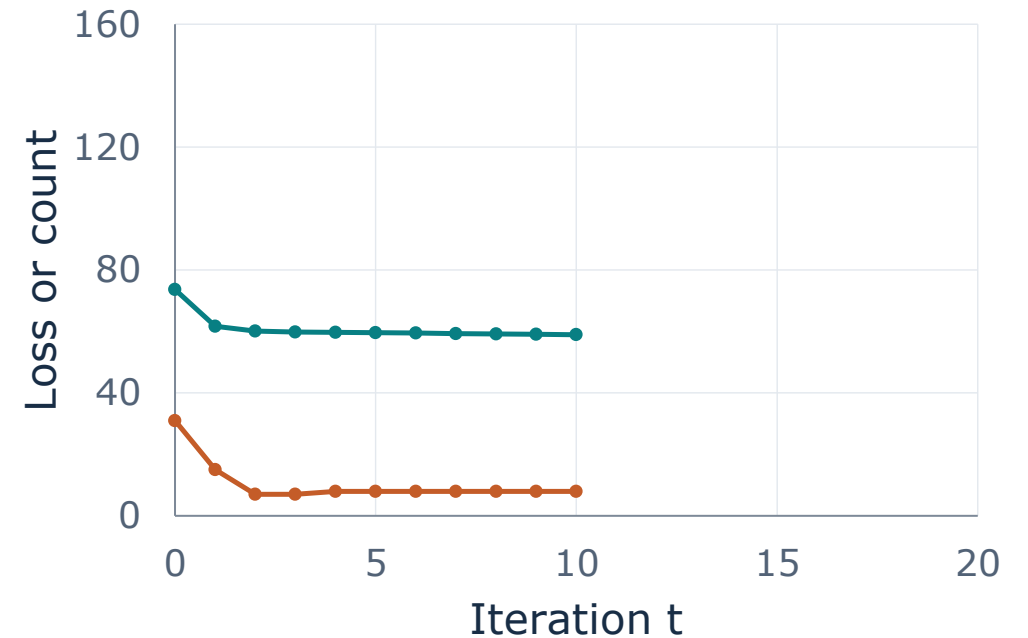
Gradient descent: $\eta = 0.001$, $t = 10$

Coefficients: $b^{(10)} \approx -5.753743$ $w_1^{(10)} \approx 0.034272$ $w_2^{(10)} \approx 1.148470$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.0099 - 0.0298x_1$$

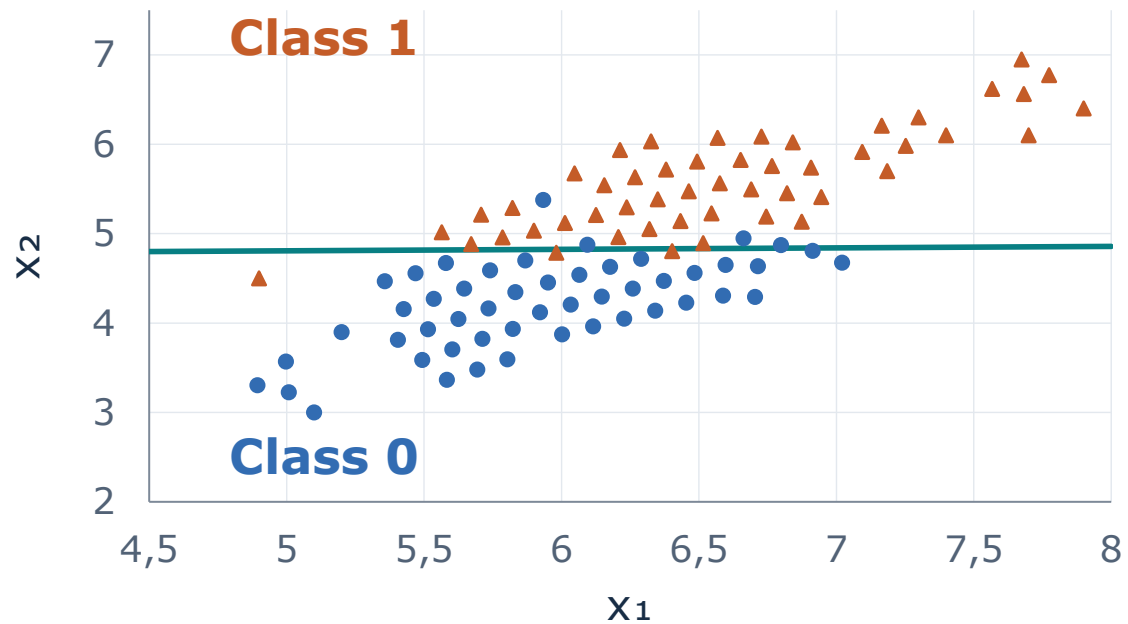


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 40.848
Mistakes: 8 of 100

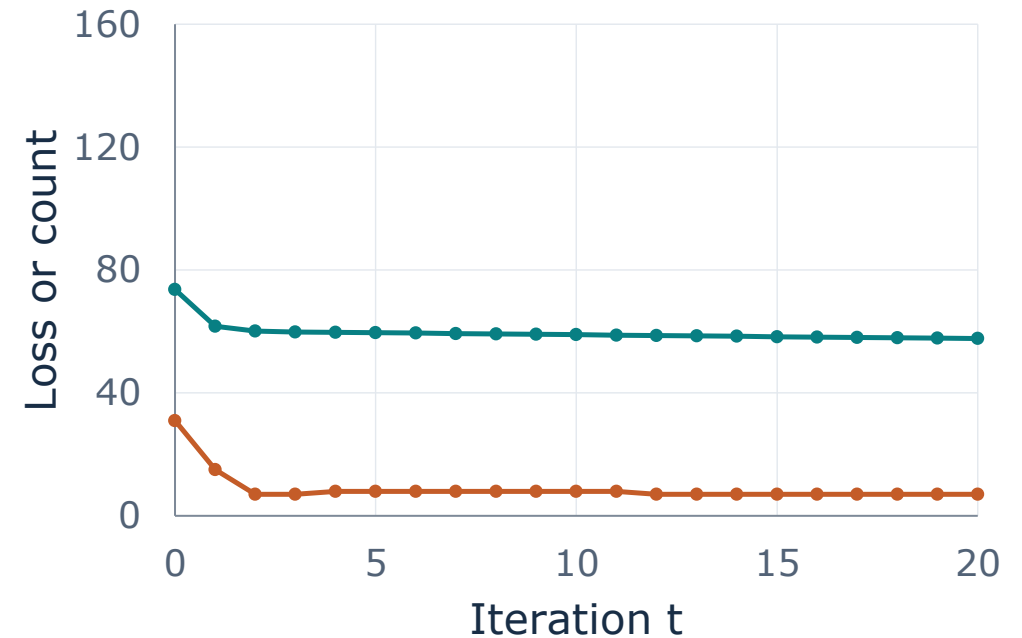
Gradient descent: $\eta = 0.001$, $t = 20$

Coefficients: $b^{(20)} \approx -5.771210$ $w_1^{(20)} \approx -0.020164$ $w_2^{(20)} \approx 1.221302$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.7255 + 0.0165x_1$$

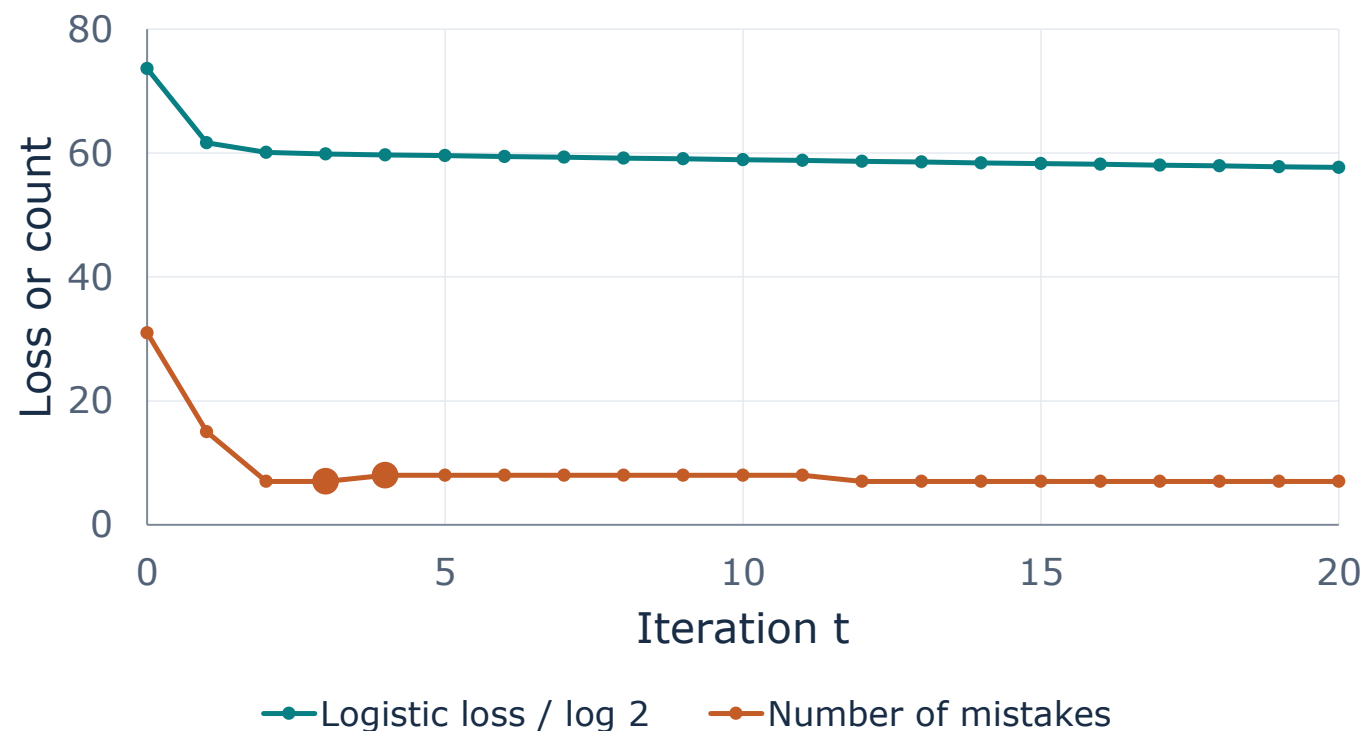


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 39.992
Mistakes: 7 of 100

The loss can fall while mistakes increase

The same planar run shows the limitation of the surrogate.



Step	Logistic loss	Errors
3	41.4948	7
4	41.3932	8

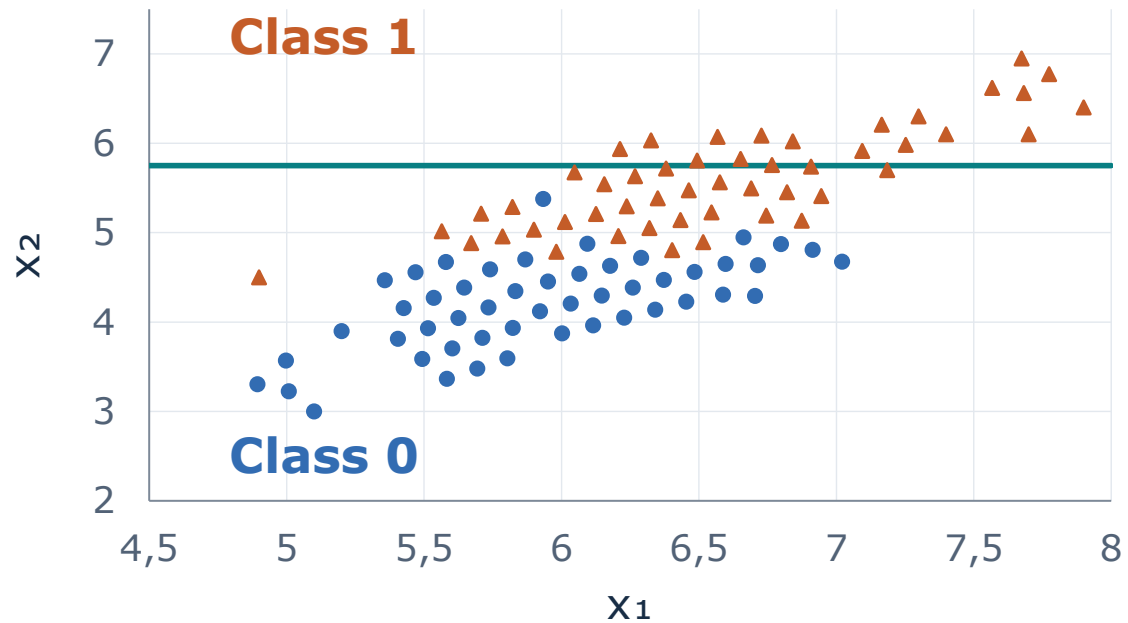
The bound still holds at both parameter choices.

Logistic loss can decrease even when the mistake count increases.

Gradient descent: $\eta = 0.002$, $t = 0$

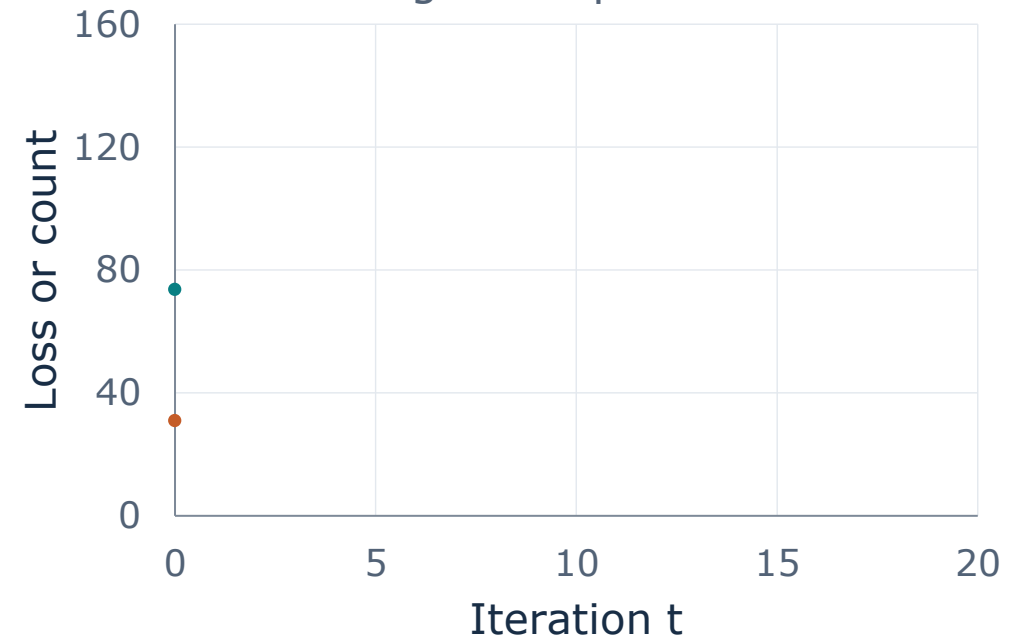
Initial coefficients: $b^{(0)} = -5.75$, $w_1^{(0)} = 0$, $w_2^{(0)} = 1$

Learning rate: $\eta = 0.002$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.7500 + 0.0000x_1$$

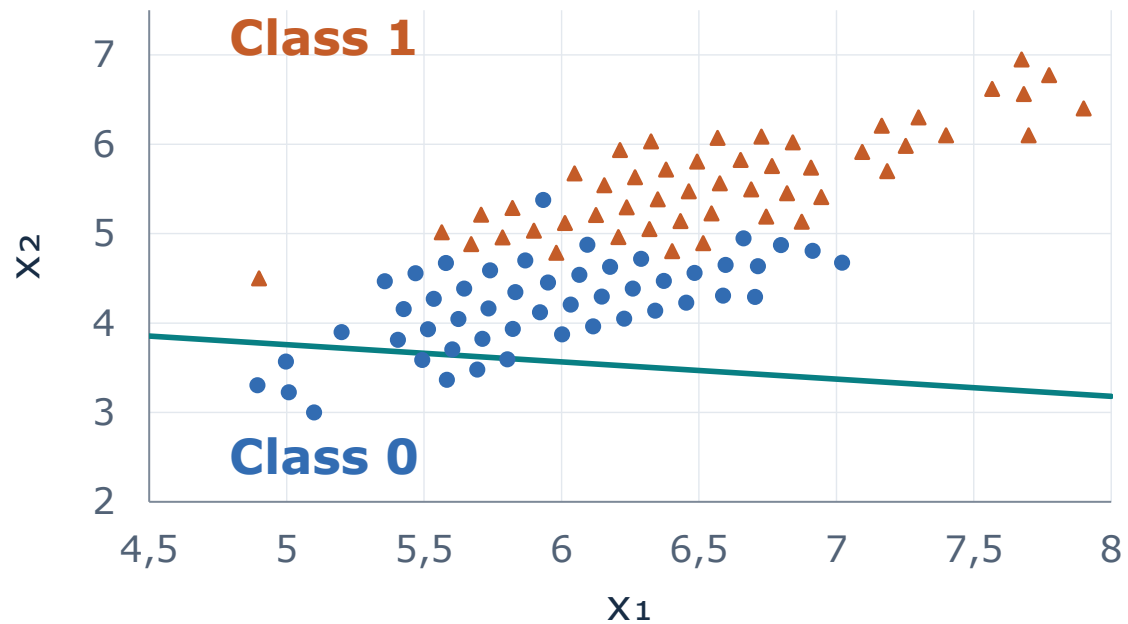


—●— Logistic loss / log 2 —●— Number of mistakes

Logistic loss: 51.048
Mistakes: 31 of 100

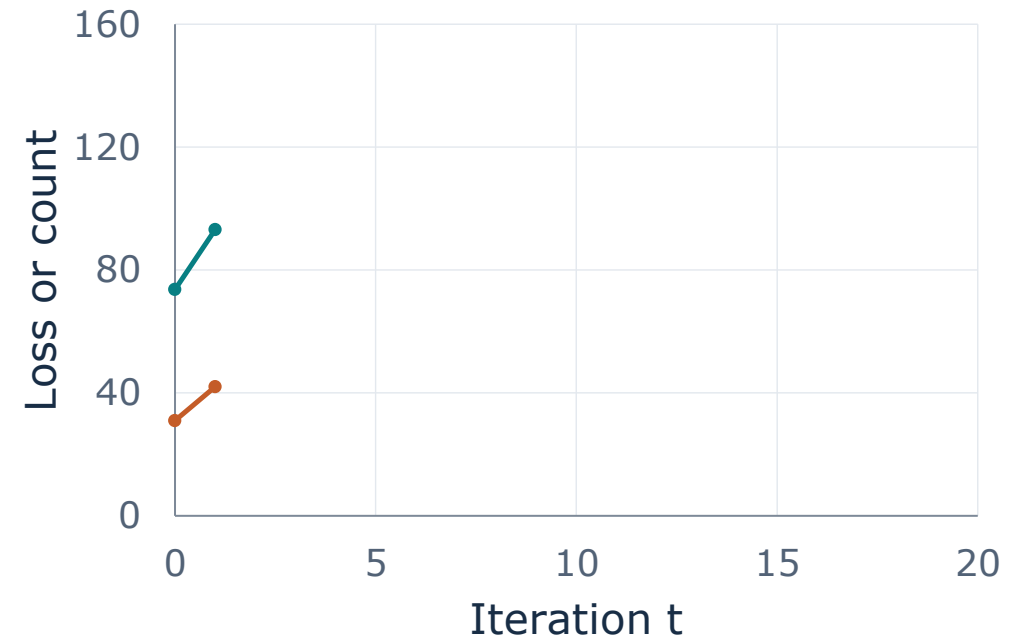
Gradient descent: $\eta = 0.002$, $t = 1$

Coefficients: $b^{(1)} \approx -5.715164$ $w_1^{(1)} \approx 0.233238$ $w_2^{(1)} \approx 1.210411$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.7217 - 0.1927x_1$$

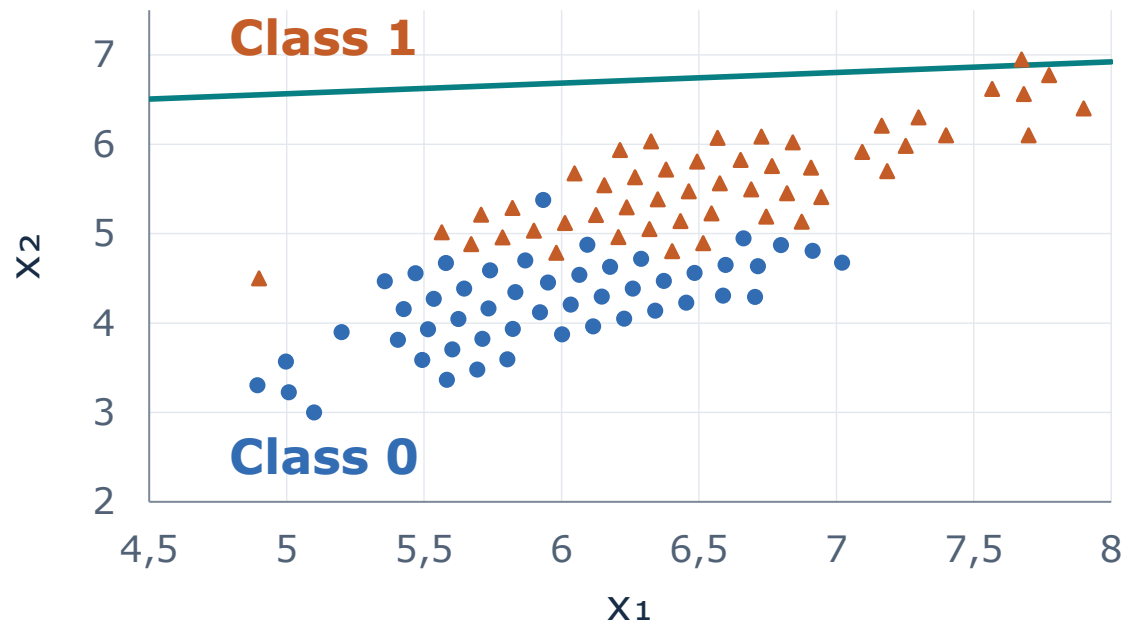


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 64.556
Mistakes: 42 of 100

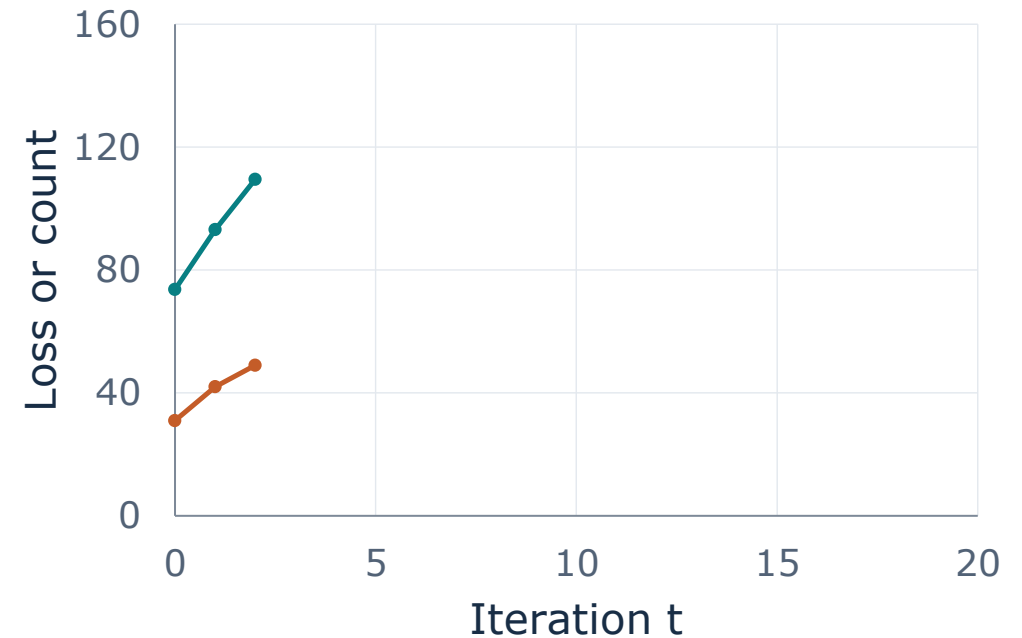
Gradient descent: $\eta = 0.002$, $t = 2$

Coefficients: $b^{(2)} \approx -5.773416$ $w_1^{(2)} \approx -0.115049$ $w_2^{(2)} \approx 0.966988$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.9705 + 0.1190x_1$$

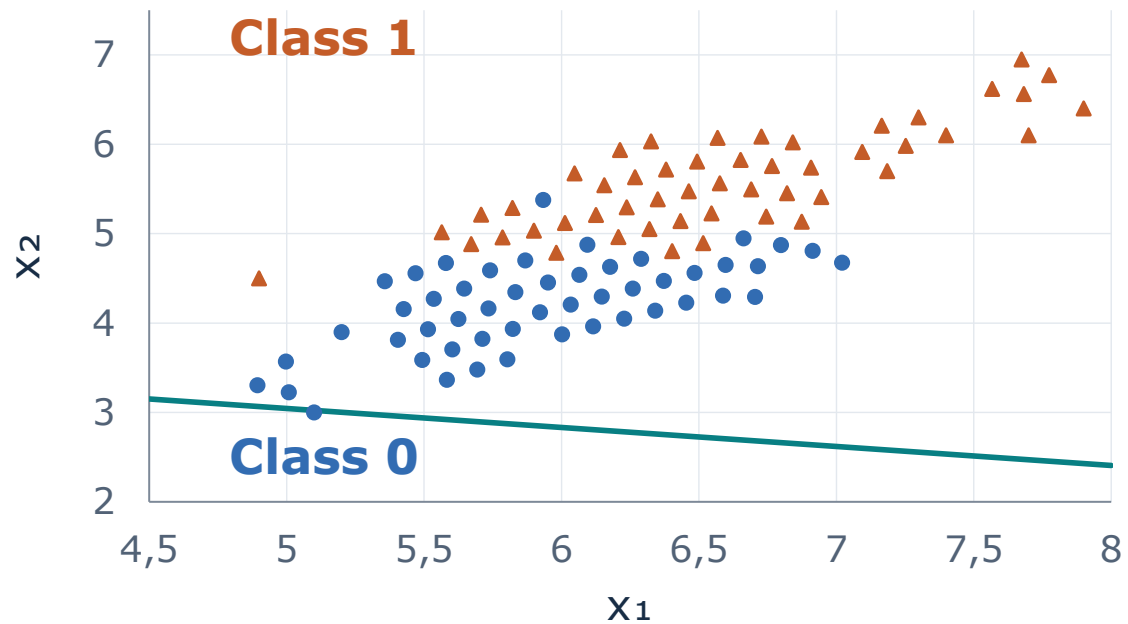


—●— Logistic loss / log 2 —●— Number of mistakes

Logistic loss: 75.928
Mistakes: 49 of 100

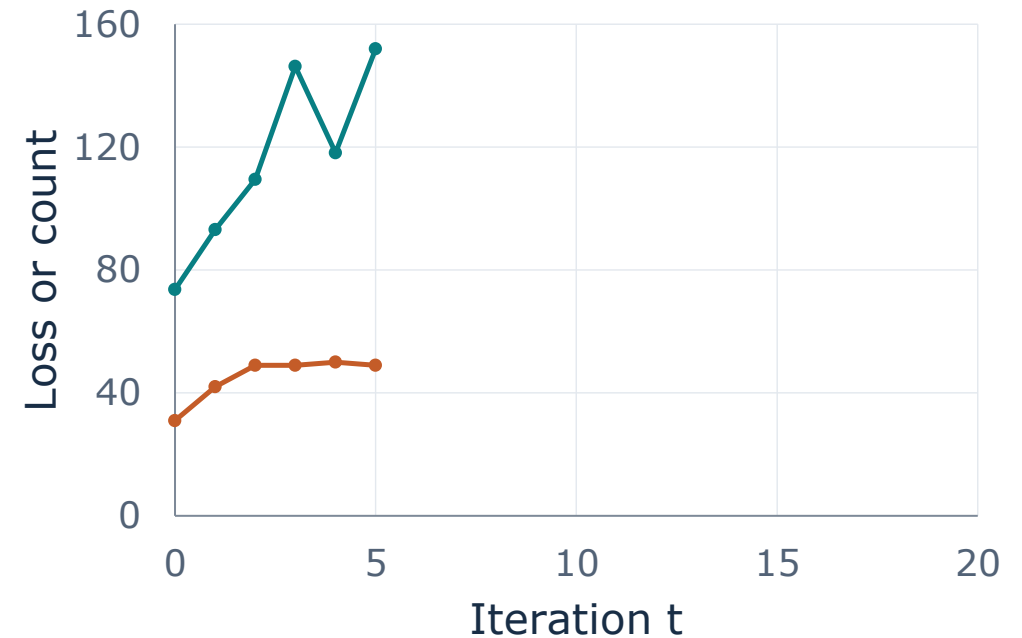
Gradient descent: $\eta = 0.002$, $t = 5$

Coefficients: $b^{(5)} \approx -5.718821$ $w_1^{(5)} \approx 0.295982$ $w_2^{(5)} \approx 1.392324$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.1074 - 0.2126x_1$$

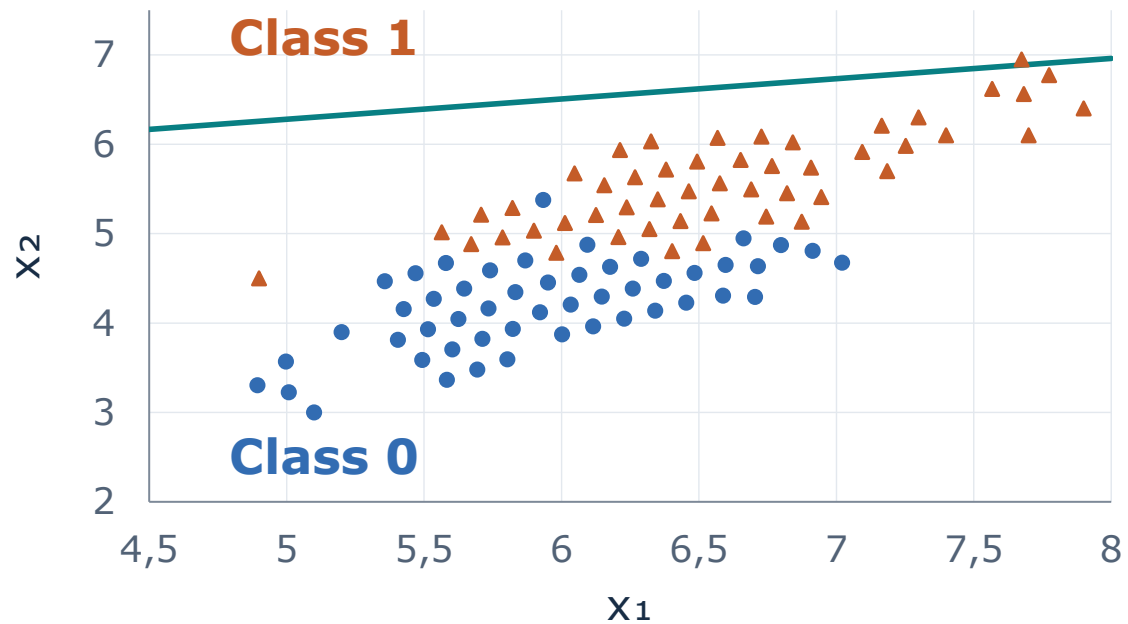


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 105.343
Mistakes: 49 of 100

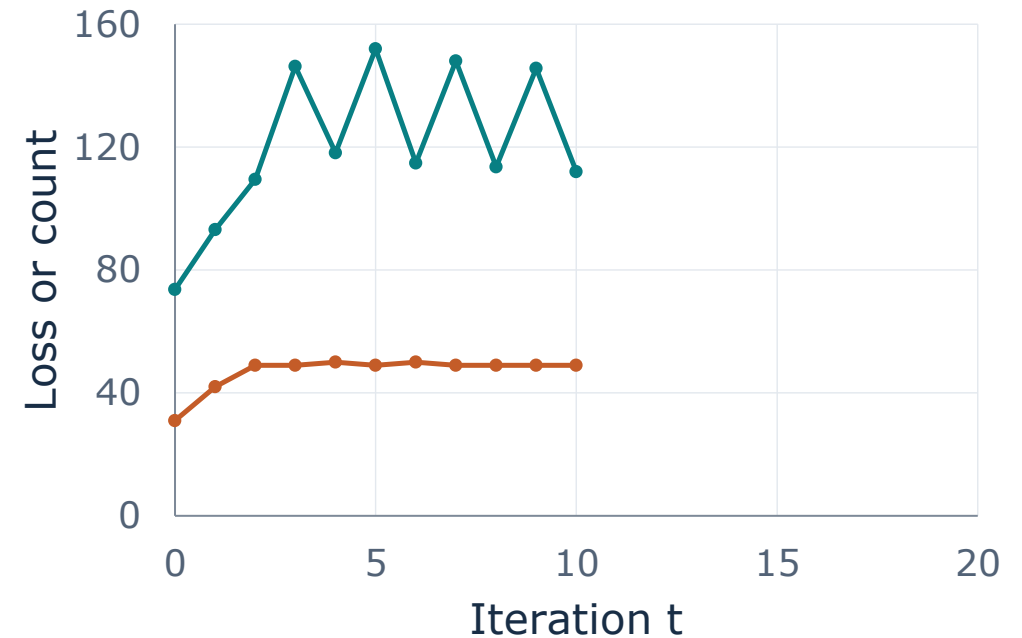
Gradient descent: $\eta = 0.002$, $t = 10$

Coefficients: $b^{(10)} \approx -5.825631$ $w_1^{(10)} \approx -0.256765$ $w_2^{(10)} \approx 1.132075$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.1460 + 0.2268x_1$$

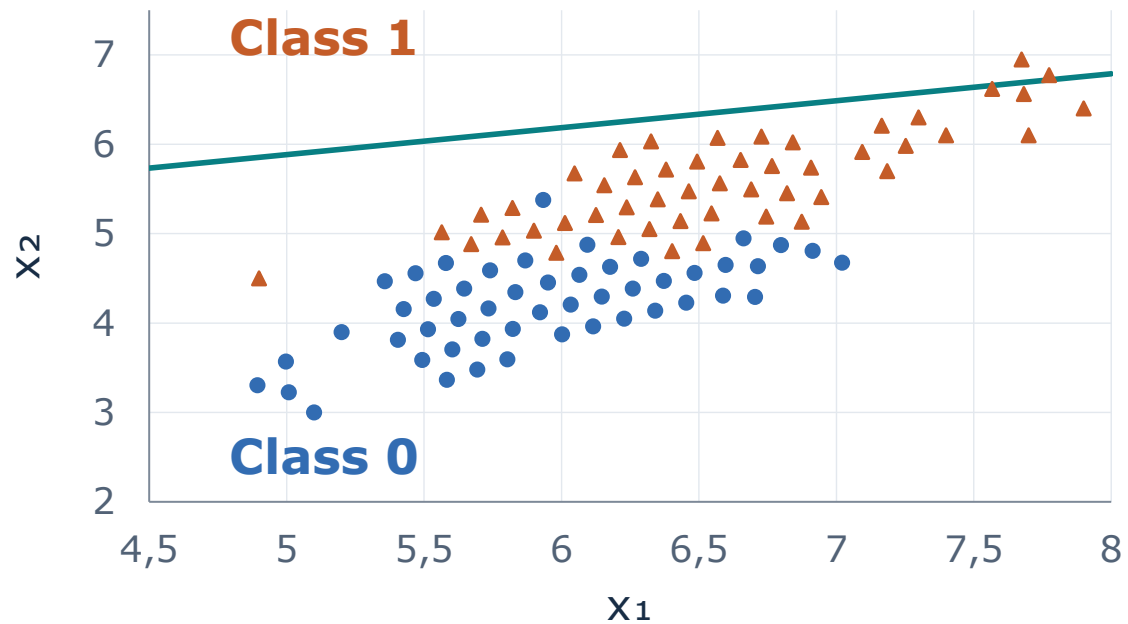


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 77.632
Mistakes: 49 of 100

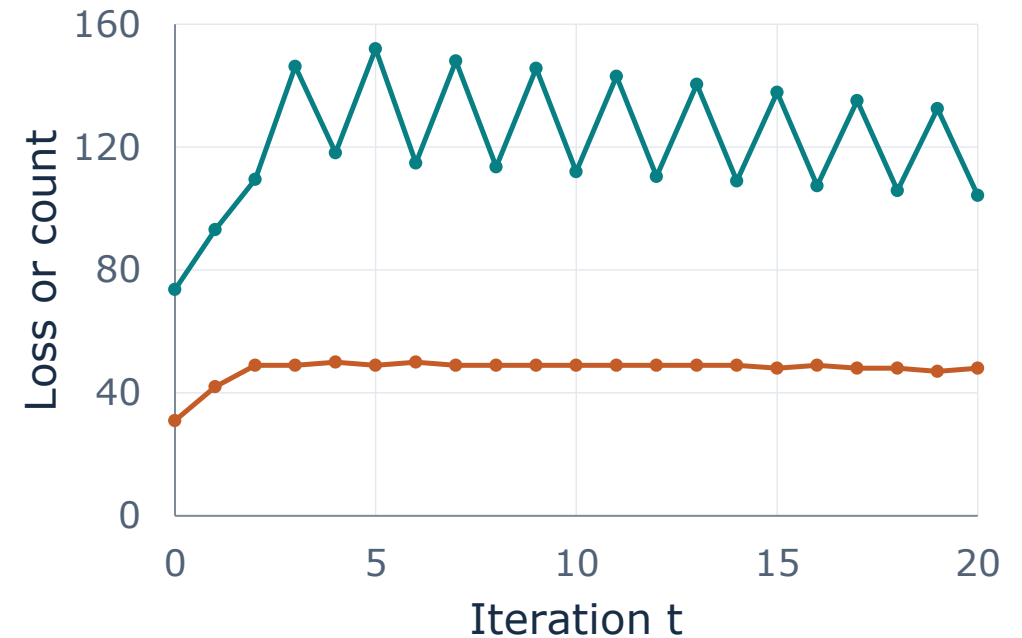
Gradient descent: $\eta = 0.002$, $t = 20$

Coefficients: $b^{(20)} \approx -5.884803$ $w_1^{(20)} \approx -0.405317$ $w_2^{(20)} \approx 1.344522$



• Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.3769 + 0.3015x_1$$

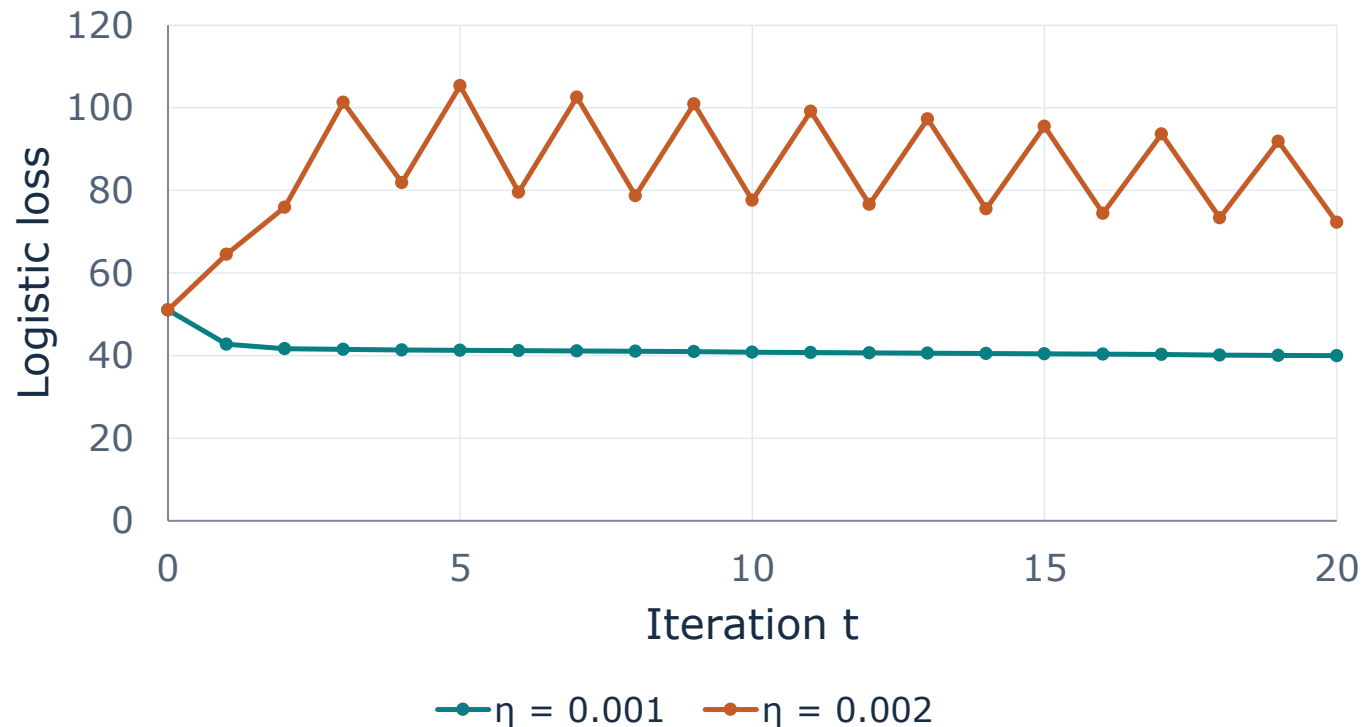


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 72.282
Mistakes: 48 of 100

Step size and logistic loss

Same data and initial coefficients; only the learning rate changes.



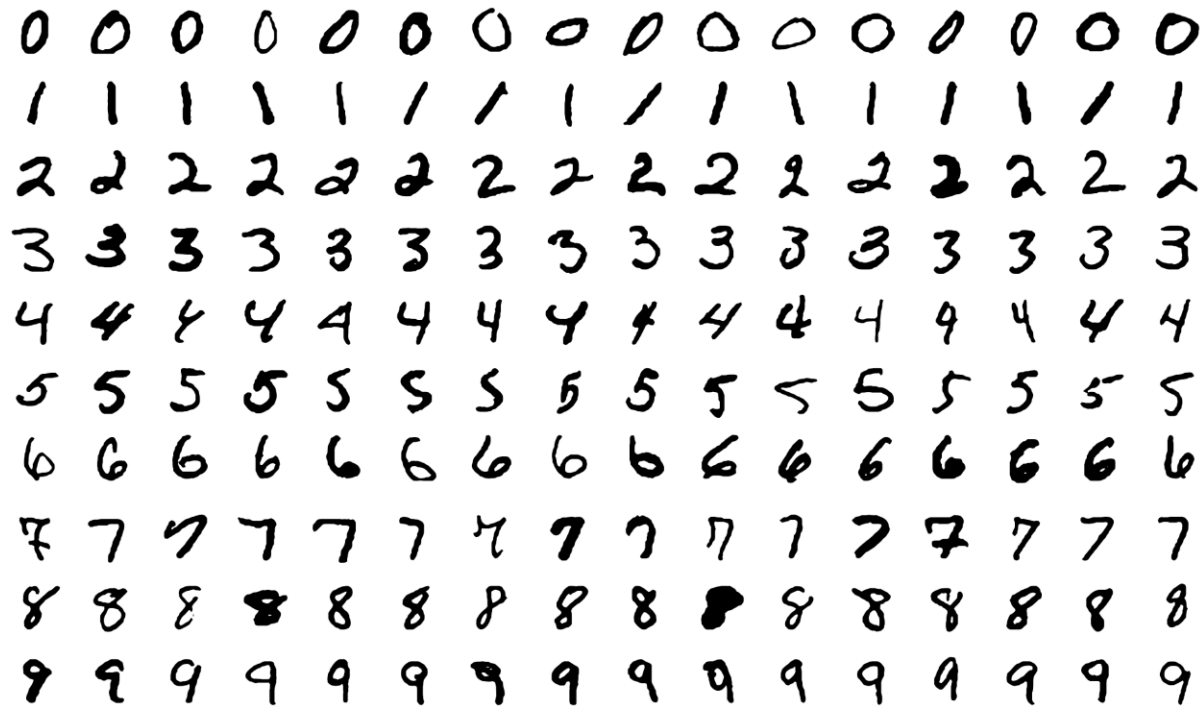
Step	Loss $\eta = 0.001$	Loss $\eta = 0.002$
0	51.048	51.048
1	42.733	64.556

**At $\eta = 0.002$,
even the first update
increases the loss.**

A step that is too large can increase the loss.

More than two classes

Recognising a handwritten digit means choosing among ten classes: 0, 1, ..., 9.



Each row contains examples of the same digit.

An image is made of pixels.
In greyscale, their intensities
form a matrix of real numbers.

We read the matrix row by row
to form a row vector.

We use the same pixel order
for every image.

$$\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^{1 \times d}$$

Here, d is the number of pixels.

We predict one probability per class and choose the most probable label.

Predicting class probabilities

A probabilistic predictor assigns one probability to each possible class $\{1, \dots, K\}$.

$$\mathbf{p}: \mathbb{R}^{1 \times d} \rightarrow [0,1]^{1 \times K}$$

$$\mathbf{p}(\mathbf{x}) = (p_1(\mathbf{x}), \dots, p_K(\mathbf{x}))$$

Each component gives the predicted probability of the corresponding class.
For this vector to represent a valid probability distribution,
we require its components to sum to one:

$$\sum_{k=1}^K p_k(\mathbf{x}) = 1$$

If we must choose one label, which should we predict?

The multiclass plug-in rule

Choose the class with the largest predicted probability.

$$h_{\mathbf{p}}: \mathbb{R}^{1 \times d} \rightarrow \{1, \dots, K\}$$

$$h_{\mathbf{p}}(\mathbf{x}) = \max(\operatorname{argmax}_{k \in \{1, \dots, K\}} p_k(\mathbf{x}))$$

argmax returns the set of indices of the largest components.
The outer $\max$ chooses the largest index (in case of a tie).

$$\begin{aligned} \mathbf{p}(\mathbf{x}) &= (0.25, 0.2, 0.12, 0.18, 0.25) \\ \Rightarrow \operatorname{argmax}_{k \in \{1, \dots, 5\}} p_k(\mathbf{x}) &= \{1, 5\} \\ \Rightarrow h_{\mathbf{p}}(\mathbf{x}) &= \max\{1, 5\} = 5 \end{aligned}$$

The largest probability need not exceed one half.

How do we build the probability function?

One affine score per class

As in binary classification, start with affine scores, one for each class.
for each class $k = 1, \dots, K$

$$b_k \in \mathbb{R}, \quad \mathbf{w}_k = \begin{pmatrix} w_{1k} \\ \vdots \\ w_{dk} \end{pmatrix} \in \mathbb{R}^{d \times 1} \quad \boldsymbol{\theta}_k = \begin{pmatrix} b_k \\ \mathbf{w}_k \end{pmatrix} \in \mathbb{R}^{(d+1) \times 1}$$

The k -th score function is defined via

$$f_{\boldsymbol{\theta}_k}(\mathbf{x}) = b_k + \mathbf{x}\mathbf{w}_k = (1, \mathbf{x})\boldsymbol{\theta}_k$$

Stack all the parameter columns in a matrix Θ

$$\Theta = (\boldsymbol{\theta}_1 \quad \dots \quad \boldsymbol{\theta}_K) \in \mathbb{R}^{(d+1) \times K}$$

$$\mathbf{f}_{\Theta}(\mathbf{x}) = \left(f_{\boldsymbol{\theta}_1}(\mathbf{x}), \dots, f_{\boldsymbol{\theta}_K}(\mathbf{x}) \right) = (1, \mathbf{x})\Theta \in \mathbb{R}^{1 \times K}$$

Softmax turns scores into probabilities

Exponentiate each score, then divide by the sum across classes.

$$\mathbf{s} = (s_1, \dots, s_K) \in \mathbb{R}^{1 \times K}$$

$$\text{softmax}: \mathbb{R}^{1 \times K} \rightarrow (0,1)^{1 \times K}$$

$$[\text{softmax}(\mathbf{s})]_k = \frac{e^{s_k}}{\sum_{r=1}^K e^{s_r}}, \quad k = 1, \dots, K$$

$$[\text{softmax}(\mathbf{s})]_k > 0$$

$$\sum_{k=1}^K [\text{softmax}(\mathbf{s})]_k = \frac{\sum_{k=1}^K e^{s_k}}{\sum_{r=1}^K e^{s_r}} = 1$$

Warning: separate sigmoids do not ensure a sum of one!!!
Three zero scores would give three probabilities of one half.

Our multiclass probability model

Apply softmax to the row of affine scores.

$$\mathbf{p}_{\Theta}(\mathbf{x}) = \text{softmax}(\mathbf{f}_{\Theta}(\mathbf{x}))$$

Its component for each class is

$$p_{\Theta,k}(\mathbf{x}) := [\mathbf{p}_{\Theta}(\mathbf{x})]_k = \frac{\exp(f_{\Theta_k}(\mathbf{x}))}{\sum_{r=1}^K \exp(f_{\Theta_r}(\mathbf{x}))}, \quad k = 1, \dots, K$$

Apply the same **plug-in rule** to these probabilities.

$$h_{\mathbf{p}_{\Theta}}(\mathbf{x}) = \max(\text{argmax}_{k \in \{1, \dots, K\}} p_{\Theta,k}(\mathbf{x}))$$

The largest score and largest probability identify the same class.

An example with three classes

Fix one input and suppose its three affine scores are as follows.

$$\mathbf{f}_{\Theta}(\mathbf{x}) = (0, \log 2, \log 3)$$

$$(e^0, e^{\log 2}, e^{\log 3}) = (1, 2, 3)$$

$$\mathbf{p}_{\Theta}(\mathbf{x}) = \left(\frac{1}{1+2+3}, \frac{2}{1+2+3}, \frac{3}{1+2+3} \right) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2} \right)$$

$$h_{\mathbf{p}_{\Theta}}(\mathbf{x}) = \max(\operatorname{argmax}_{k \in \{1,2,3\}} p_{\Theta,k}(\mathbf{x})) = \max(\{3\}) = 3$$

Class 3 has the largest predicted probability.

Cross-entropy loss for the observed class

For a probability row and an observed class, use the probability of that class.

$$\mathbf{p} = (p_1, \dots, p_K), \quad y \in \{1, \dots, K\}, \quad p_y > 0$$

$$\ell_{\text{multi}}(\mathbf{p}, y) = -\log(p_y)$$

$$\mathbf{p} = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right), \quad y = 2$$

$$\ell_{\text{multi}}(\mathbf{p}, 2) = -\log(1/3) = \log(3)$$

Using the plug-in rule, the prediction is class 3, but the observed class is 2.
The loss uses the probability assigned to class 2.

The multiclass fitting objective

For N labelled observations, add the individual losses.

$$(\mathbf{x}_i, y_i), \quad i = 1, \dots, N, \quad \mathbf{x}_i \in \mathbb{R}^{1 \times d}, \quad y_i \in \{1, \dots, K\}$$

$$\mathcal{L}_{\text{multi}}(\Theta) = \sum_{i=1}^N \ell_{\text{multi}}(\mathbf{p}_{\Theta}(\mathbf{x}_i), y_i)$$

Fit the parameter matrix by minimising this summed loss, e.g., via gradient descent.

Models and fitting losses

Binary and multiclass models turn affine scores into class probabilities.

Task	Model predictions	Fitting loss
Regression	Numerical values	Squared loss
Binary classification	Sigmoid probabilities	Logistic loss
Multiclass classification	Softmax probabilities	Cross-entropy loss

We fit probabilities with a smooth loss.

We evaluate classification by counting mistakes.

Lecture notes and exercises explain the material in more detail.

Exercise solutions will be posted before the next lecture.

github.com/semtm0051/2026 27

